



Build a smarter world

Eupfin Tracker Device ODM Service

Product Proposal

V1.8

2026.08.28



Contents

01 Customer requirements

02 Technical solution evaluation

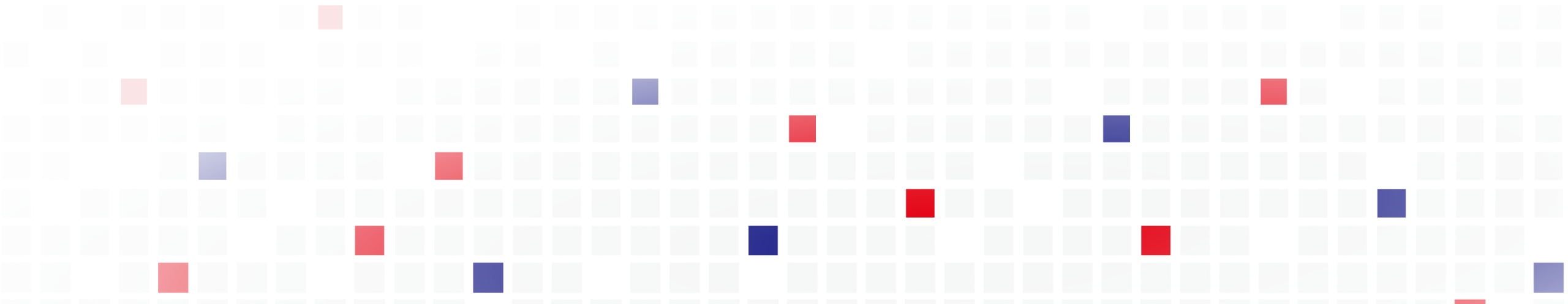
03 Project plan

04 Quotation budgeting

05 Supplementary information



01 Customer requirements



Customer Requirements—V1.8 Update

車用追蹤器客戶需求表

一、基本資訊

公司名稱*	EUPFIN TECHNOLOGY CO., LTD.	聯絡窗口*	
Email*		電話	
專案名稱*		預計量產時間	2026Q4

二、應用場景

應用類型*	車隊管理	車隊管理 / 防盜 / 冷鏈 / 租賃車 / 工程車 / 其他	車種*	小客車 / 貨車 / 巴士 / 拖車 / 其他	機車 / 小客車 / 貨車 / 巴士 / 拖車 / 其他
安裝方式*	有線式/OBD	有線式 / OBD / 電池式 / 太陽能 / 其他	使用國家/地區*	Taiwan、Vietnam、Malaysia、Thailand、Indonesia	
是否跨國使用*	單國&漫遊	單國 / 漫遊	使用環境	戶外、高震動、長時間停車	戶外 / 高溫 / 高震動 / 潮濕 / 長時間停車 / 其他

三、主要需求

主要功能需求*	即時定位、歷史軌跡、ACC、斷電告警、溫度監控	即時定位、歷史軌跡、電子圍籬、ACC、斷電告警、溫度監控...	平台需求*	只要硬體 / 需要整合支援	只要硬體 / 要 API / 要平台 / 需要整合支援
希望通訊技術	2G、3G、4G頻段	2G / LTE Cat.1 / Cat.1 bis / LTE-M / NB-IoT / 5G / 不確定	是否需備援電池	可支援備援電池	是 / 否
是否需 OTA / 遠端設定	是	是 / 否	認證需求	Vietnam : QCVN55、QCVN117 Thailand : NBTC Malaysia : SIRIM Taiwan : NCC、BSMI	CE / FCC / NCC / E-mark / 其他
SIM卡需求	翻蓋式固定、nano-sim card、需熱插拔功能	插卡式 或 eSIM 需求	電壓需求	9~36V	
天線	內置	內置 or 外置	尺寸需求		

四、數量與商務

樣品數量*	5 pcs	預估年度需求量	100kpcs/1yr, totally 5yrs
預估量產數量*		客製需求	Logo / 包裝 / 韌體 / 線材 / 外殼 / 無
預算區間			

Customer Requirements

車用追蹤器客戶需求表

備註 / 其他特殊需求

- 1. 2組RS-232介面
- 2. 支援RFID功能，工作頻率13.56MHz，符合ISO/IEC 15693標準
- 3. 1組DB9介面(Baud rate：115200 bps)
- 4. 4組Input
- 5. 1組Output (Floating / High Impedance) 負輸出
- 6. motion sensor (6-axis accelerometer + gyroscope)
- 7. 1組1-W 溫度介面(可偵測4組溫度棒)
- 8. 足夠記憶體空間，儲存至少30天資訊
- 9. 燈號：網路燈號、GPS燈號、Memory燈號、駕駛者燈號
- 10. 警報器(蜂鳴器)
- 11. 藍芽功能
- 12. CAN BUS功能

Reference Device



Operation Environment			
Temperature	Activities	-20÷85°C ±0.5%	
	Storage	0~60°C	
humidity		5÷90% ±2.5%	
Electrical characteristics and consumption			
Active range		9÷36VDC	
Rated supply voltage		12/24VDC	
Power Consumption	Active	48÷60mA	@12VDC
	Idle	22÷24mA	
	GPS sleep	10÷12mA	
Protective function		Anti- short circuit, over current	
		Anti- over voltage, elimination of voltage spikes	
		Reverse polarity	
Backup power		Battery: 3.7V – 150mAh (HHS 402025 150mAh 0.555Wh)	Certificate: UN38.3 1.2M Drop test MSDS report
Digital output		Maximum load current: 50mA Maximum voltage: 24VDC	
Digital input		0-36VDC Logic threshold: 7V	
Physical specification			
Size		94.0 x 55.0 x 16.8 (mm)	
Weight		200 Grams	
Housing		ABS + PC plastic, non-flammable	
Connector		Molex Micro-Fit3.0, 2x3 pin	
Antenna		Built- in	

Reference Device

SIM card		Nano SIM Card	
Wireless and satellite connectivity			
4G-LTE	Modem	SIMCOM A7672S	A7672S (Simcom – China)
	Antena	INTERNAL	
	Frequency band	TDD-LTE B34/B38/B39/B40/B41	
		FDD-LTE B1/B3/B5/B8	
		GSM/GPRS/EDGE 900/1800 MHz	
	Data transfer	LTE CAT1: 10(DL)/5(UL) GPRS/EDGE: 236.8(DL)/236.8(UL)	
GNSS	SMS	Available	ATGM336H5N-31 (HangZhou ZhongKe Microelectronics Co., Ltd – China)
	Protocols	TCP/IP	
	Module	ATGM336H-5N-31	
	Chipset	AT6558	
	Systems	GPS+Glonass+Beidou	
	Performance	Tracking: -162 dBm Recapture sensitivity: -160 dBm Cold Start: -148 dBm Recapture TTFF: ≤1s Hot start TTFF: ≤1s Cold Start TTFF: <35s	
RFID reader	Antenna	Built-in patch antenna ceramic with A high gain LNA	
	Accuracy	< 10m	
	Chipset	CR95HF	
	Frequency	13.553 ÷ 13.567 MHz	
	Standard	ISO/IEC 14443 Type A and B ISO/IEC 15693 (single or double subcarrier) ISO/IEC 18092	
	Main reflex	≤ 4.5 mW ERP	

	Spurious emissions	- Operating mode: 0.67 μA/m (-3.5 dB μA/m) - Standby mode: 0.06 μA/m (-24.5 dB μA/m)	
	Reading distance	Max 3cm	
	Compatible card	I.CODE SLI, TAG_IT	
PROCESSOR AND STORAGE			
CPU	Core	Arm® 32-bit Cortex®-M0+ 60 MHz	HT32F52367 (Holtek – Taiwan)
	Low-power	Available	
	Internal Flash	256 KB	
	SRAM	32KB	
	HW unit	Serial Communication interfaces	
		NVIC	
		CRC	
		DMA	
		RTC	
	External Flash	8-32MB	EN25QH64A (ESMT– Taiwan) W25Q64JWSSIQ (Winbond – Taiwan) GD25Q64ESIGR (GigaDevice – China)
Sensor (integrated on board)			
Acceleration sensor		3 axes of linear acceleration Measuring range 2÷16g high resolution 16bits	
Voltage Sensors		Input voltage Measurement	
Signal connection port			
Port A	VDC power supply	9-36VDC Power supply to the device	ACC
		1 high positive signal level	

Reference Device

	2 x Digital inputs	1 signal with configurable active level	ALTO
	1 x RS232	1 RS232	TX0, RX0
Port B	1-WIRE	1-WIRE data communication	1-WIRE
	1 x Digital input	1 signal with configurable active level	ALT1
	1 x Digital Output	Low positivity level with trigger and pulse type support	Dout
	1 x RS232	1 RS232	TX1, RX1
	1 x Analog input	Analog signal input voltage range 0-33V	
01 x COM (DB9 Male)		Configuration port and data extraction	
Status Indicators			
Led light		4 LED: DRI, GPS, CELL, MEM	
Buzzer		Available on the circuit	
Feature			
Basic	2-way confirmation handshake		
	Real-time positioning and monitoring of journeys		
	Identify driver via RFID tag		
	Fuel monitoring		
	Picture monitoring		
	Temperature monitoring		
	Digital output format: nPulse/Trigger/Level		
	Wave break detection		
	Configure and test via SMS/TCPIP		
Data storage and retransmission		30 days	
Time Sync.		NTP, GPS satellites	
Advanced		Warning has output response at Out0	

	Parallel connection of 2 fuel sensors	
	External power voltage warning	
	Integrating different types of sensors according to the Customer's requirements	
	Remote Firmware Update (OTA)	
	Support backup source	

Table 2. SPECIFICATIONS

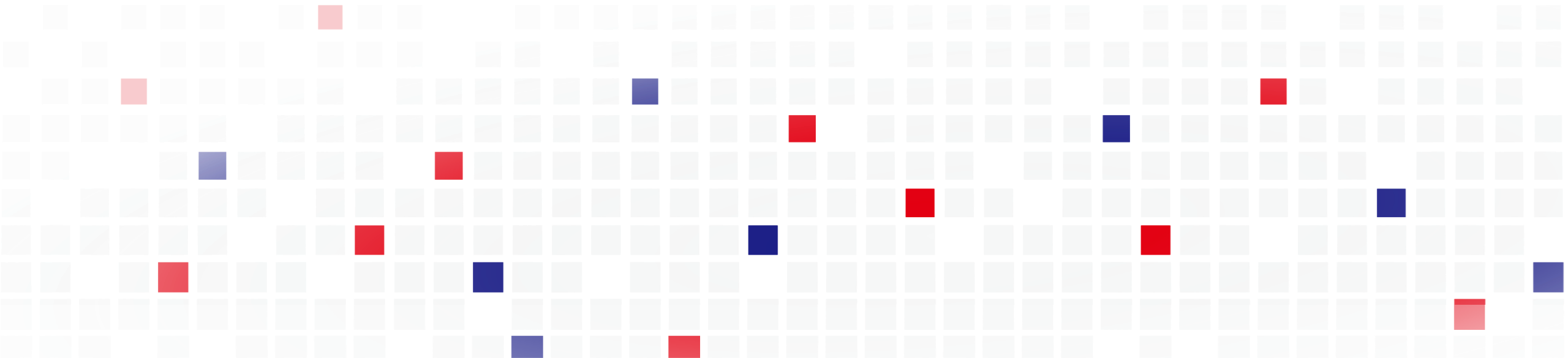
4. Port signal, and Led indicator

4.1. Led indicator

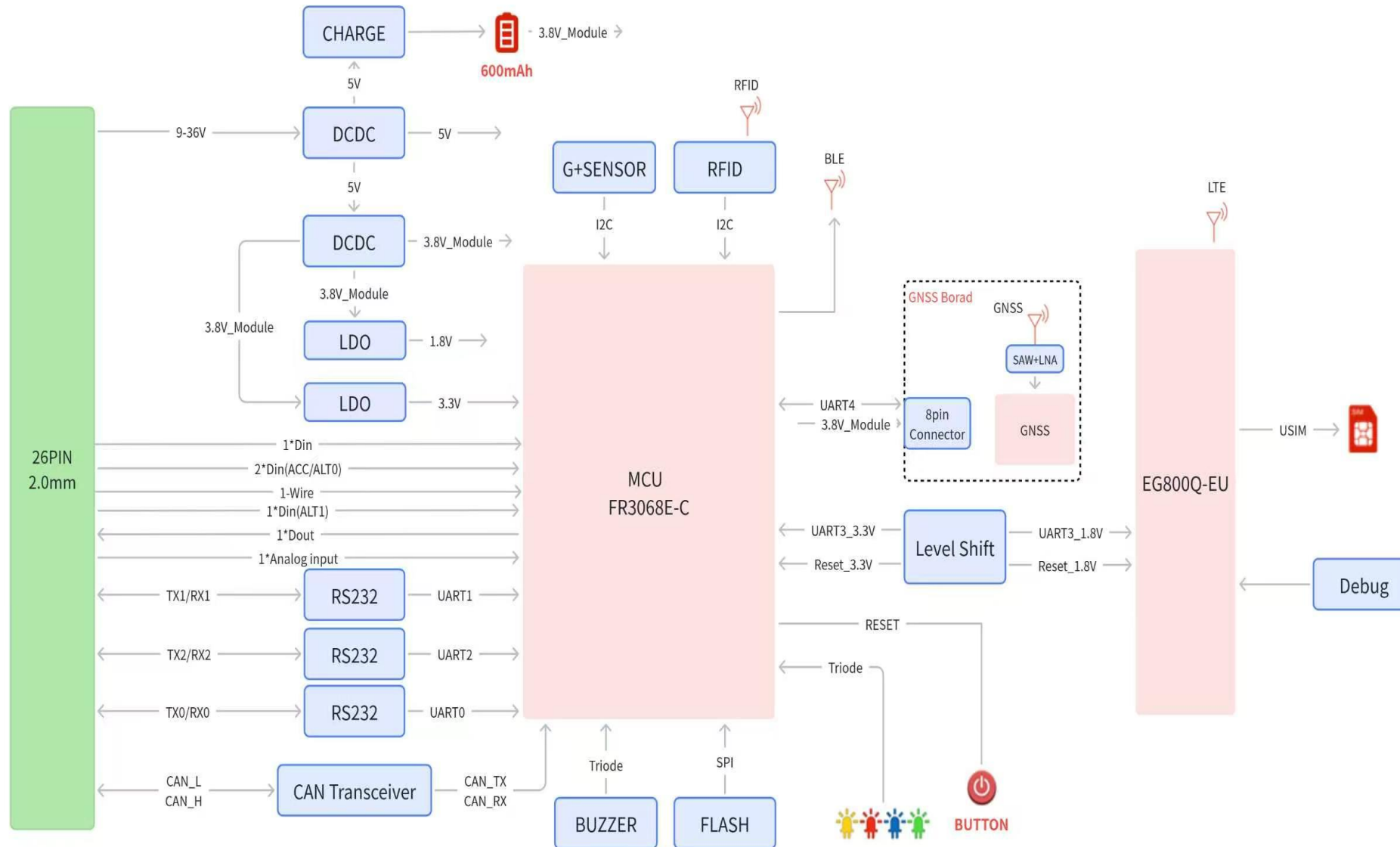
LED	Status	Descriptions
DRI (Driver)	Stable light does not flash	The driver has logged in
	Not bright	No drivers are entering
	Flashes slowly (**)	Car runs without driving log
	Flashes fast (*)	Card reader has proplems
GPS	Stable light does not flash	Tuning
	Not bright	The GNSS module has encountered a proplem
	Flashes fast (*)	In a state of sleep
	Flashes slowly (**)	Good signal, can update real- time location
CELL (CELLULAR Network)	Stable light does not flash	Registered to a data network and connecting to data server
	Not bright	Network not registered yet No sim installed/no sim accepted There was an unusual incident
	Flashes slowly(**)	Server connected successfully
MEM (Memory)	Stable light does not flash	Normal memory
	Not bright	Error/ malfunction memory
	Flashes fast	Accessing memory

Table 3.Description of LED indicator

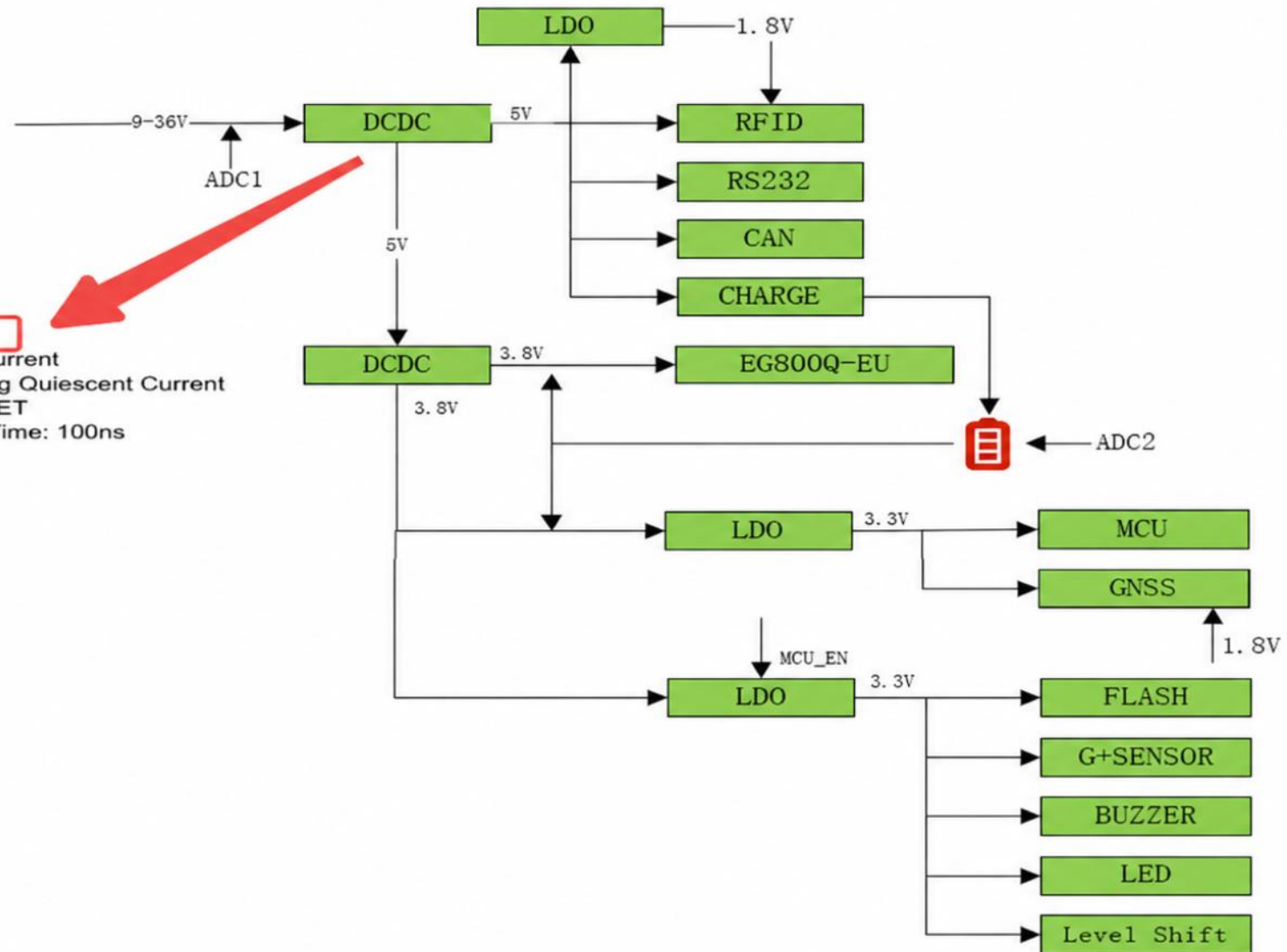
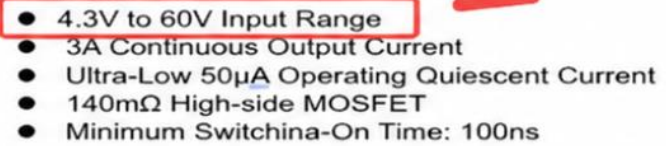
02 Technical solution evaluation



Overall design — System block diagram— V1.8 update



Overall design – System block diagram



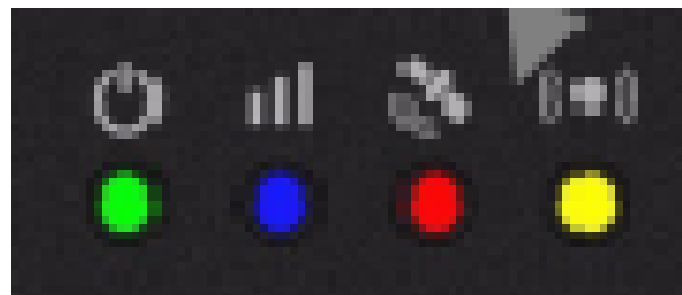
Overall design — Pin Definition—V1.8 update

PIN	讯号	颜色	规格
1	正电 B+	红	保险丝 1A
2	负电 GND	黑	—
3	ACC	蓝	数位, > 5V触发, 保险丝 1A
4	ADC	绿	类比, 0-33V
5	1-Wire	白	DS18B20 (最多4组)
6	GND	黑	GND
7	Input1	青	数位, > 5V高电位触发
8	Input2	紫	数位, > 5V高电位触发
9	Input3	棕	数位, > 5V高电位触发
10	Output	灰	负输出 (断开/GND)
11-14	RS232_1	黄/绿/黑/红	4.2mm 4Pin, 9600 bps (TX/RX/GND/ACC)
15-18	RS232_2	黄/绿/黑/红	4.2mm 4Pin, 9600 bps (TX/RX/GND/ACC)
19-22	CAN	黄/绿/黑/红	3.0mm 4Pin, J1939 (CAN_H/CAN_L/GND/ACC)
23-26	RS232_0	黄/绿/黑/红	4.2mm 4Pin, 115200 bps (TX/RX/GND/ACC, 额外加DB9公头)

Overall design – LED Status

1. Driver 黃LED :
 - a. LED恆亮：駕駛者登入
 - b. LED熄滅：駕駛者登出
2. GPS 紅LED :
 - a. LED恆亮：GPS已定位
 - b. LED熄滅：GPS未定位
3. Net 藍LED :
 - a. LED恆亮：有網路訊號且服務器連接成功
 - b. LED熄滅：無網路訊號
 - c. LED閃爍：有網路訊號
4. Power/Memory 綠LED :
 - a. LED恆亮：設備開機 且 Memory 正常
 - b. LED熄滅：設備關機/休眠
 - c. LED閃爍：設備開機 且 Memory 異常

*閃爍：1秒恆亮1秒熄滅



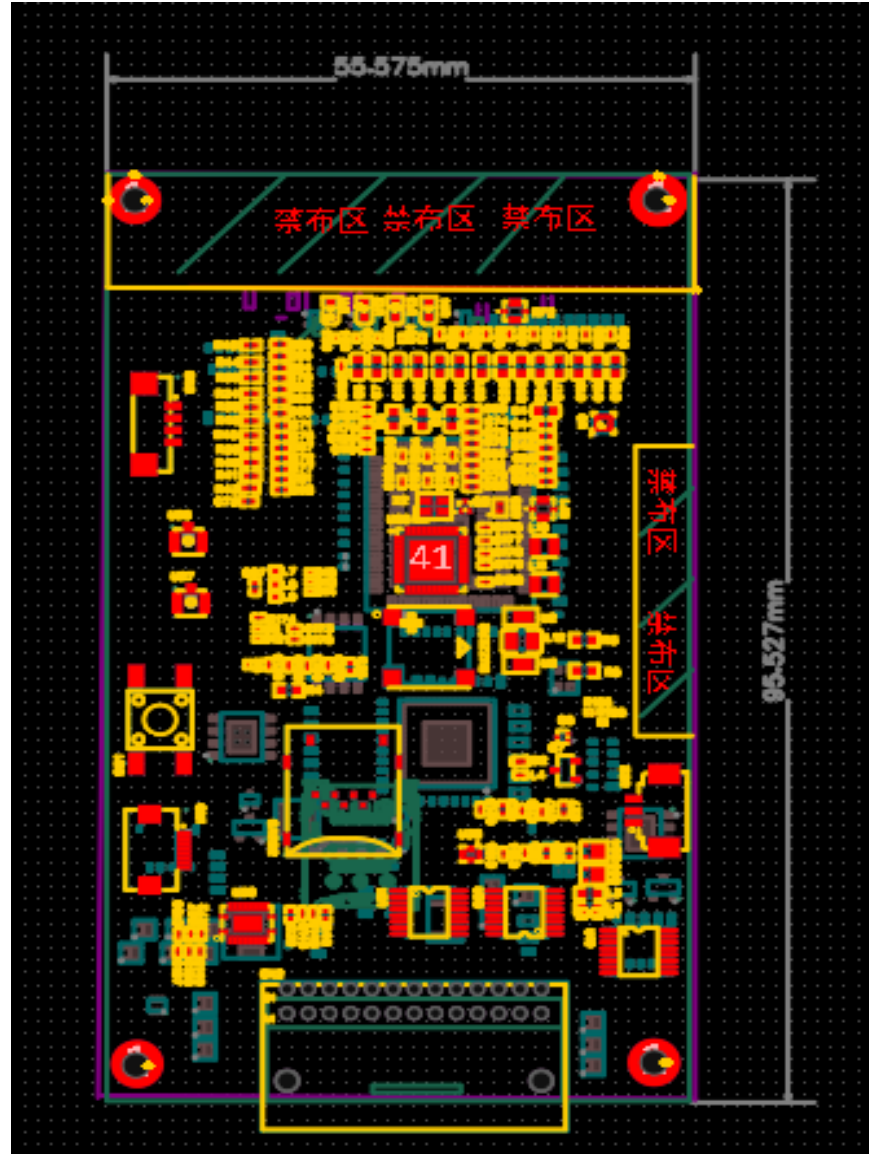
Overall design – Power Consumption Evaluation

Device Basic Configuration & Module Current Specifications					
Parameter Name	Parameter Value	Unit	Remarks		
Battery Voltage	3.7	V	Nominal battery voltage		
Battery Capacity	600	mAh	Rated battery capacity		
Discharge Efficiency	70%	%	Practical battery discharge derating factor		
Reporting Frequency	240	times/hour	Periodic data upload interval		
Working Voltage	3.3	V	Nominal voltage for digital modules		
Module	Standby Current (mA)	Operating Current (mA)	Active Duration (S)	Remarks	
LTE Module	4.5	1	0	Start up	
		30	0	Searching network	
		500	2	Send data	
GNSS	0.3	72	4	Hot start positioning	
MCU/BLE	0.1	30	1	Main control unit	
BMS	0.0007	0	0	Quiescent Current	
Function (Module)	Mode (Operating Mode)	Active Duration (min) (Single Working Duration)	Power Consumption (mW) (Instantaneous Power Consumption)	Total Power Consumption (mW) (Hourly Average Power Consumption)	
LTE module	Start up	0.0000	3.7	0.000	
LTE module	Searching network	0.0000	111	0.000	
LTE module	Send data	0.0333	1850	246.667	
LTE module	Sleep	59.9667	16.65	16.641	
GNSS	Working	0.0667	237.6	63.360	
GNSS	Sleep	59.9333	0.99	0.989	
MCU/BLE	Working	0.0167	99	6.600	
MCU/BLE	Sleep	59.9833	0.33	0.330	
BMS	Quiescent Current	60.0000	0.002534247	0.003	
Total Hourly Average Power Consumption of the Whole Device				371.8	mW
Standby Power Consumption				18.0	mW
Total Available Battery Energy				1554.0	mWh Calculated: 3.7V × mAh
Actual Standby Time (80% Discharge Efficiency)				4.2	Hours Practical application scenario

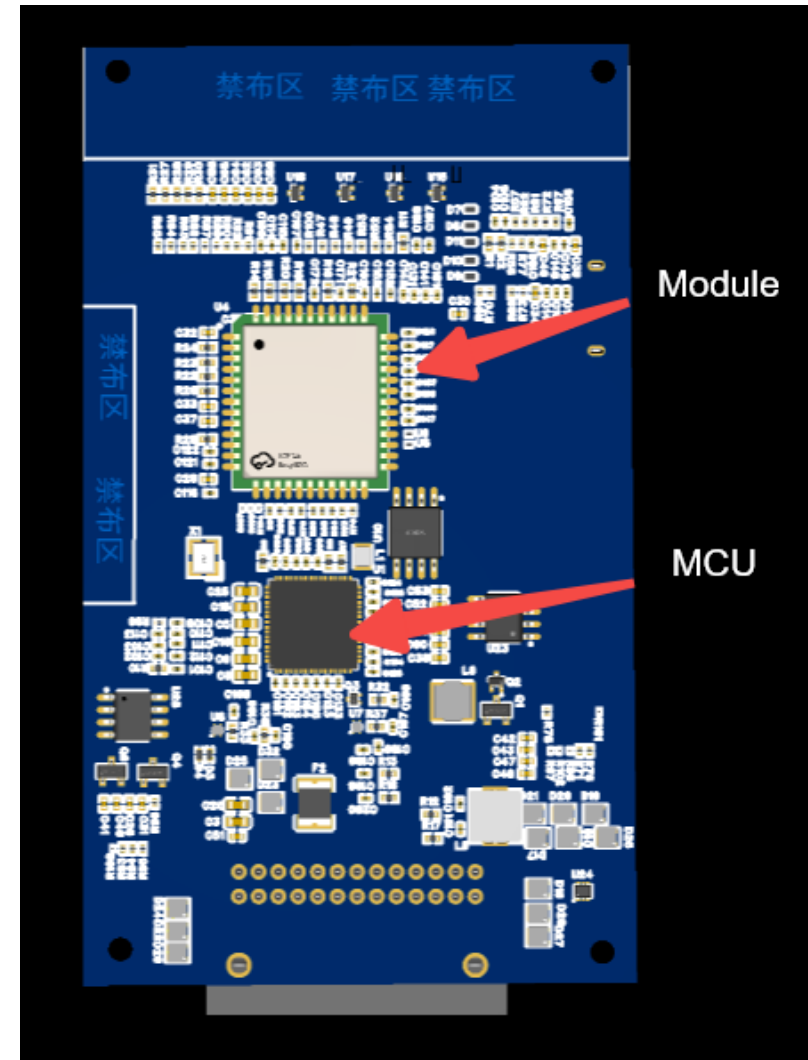
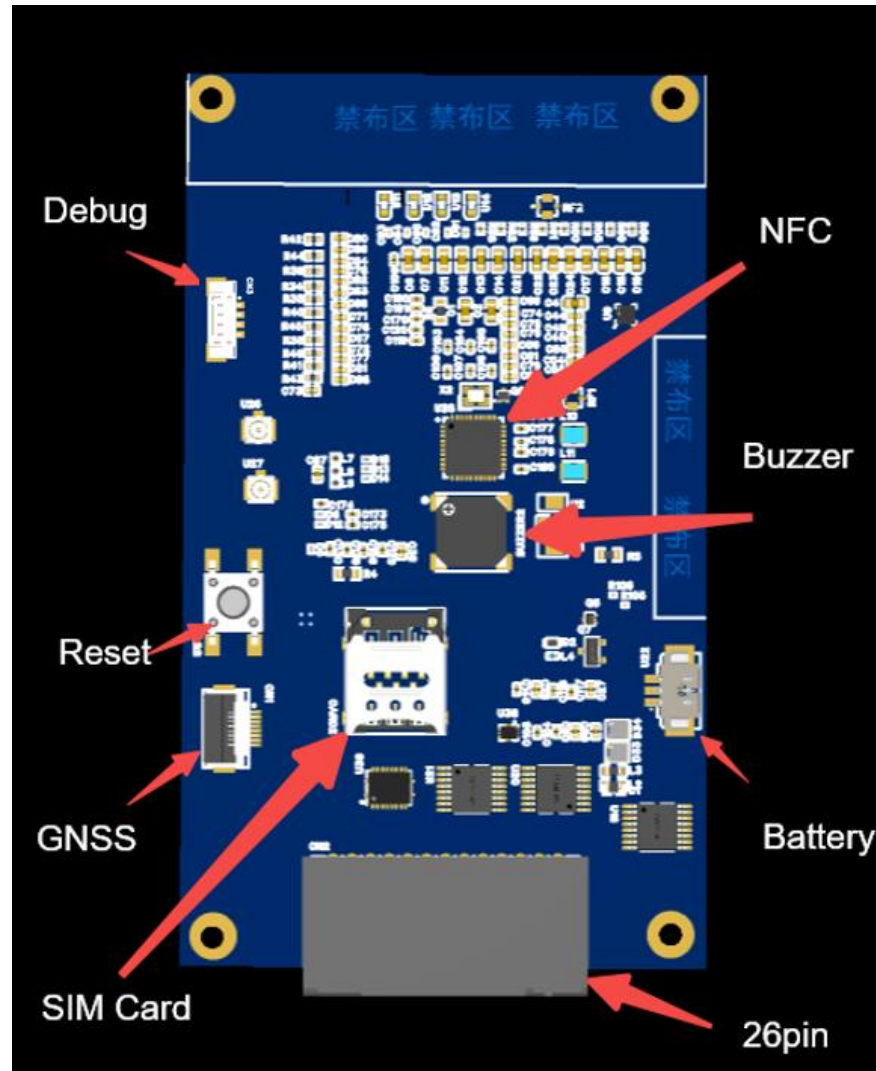
In battery-only standby mode, BLE and GNSS transmit 240 times per hour. With a 600 mAh battery and 70% discharge efficiency, the standby time is **4.2 hours**.

Overall design – Component Placement Evaluation—V1.8 update

Estimated PCBA dimensions: 96×55×1.6 mm, 4-layer through-hole PCB.

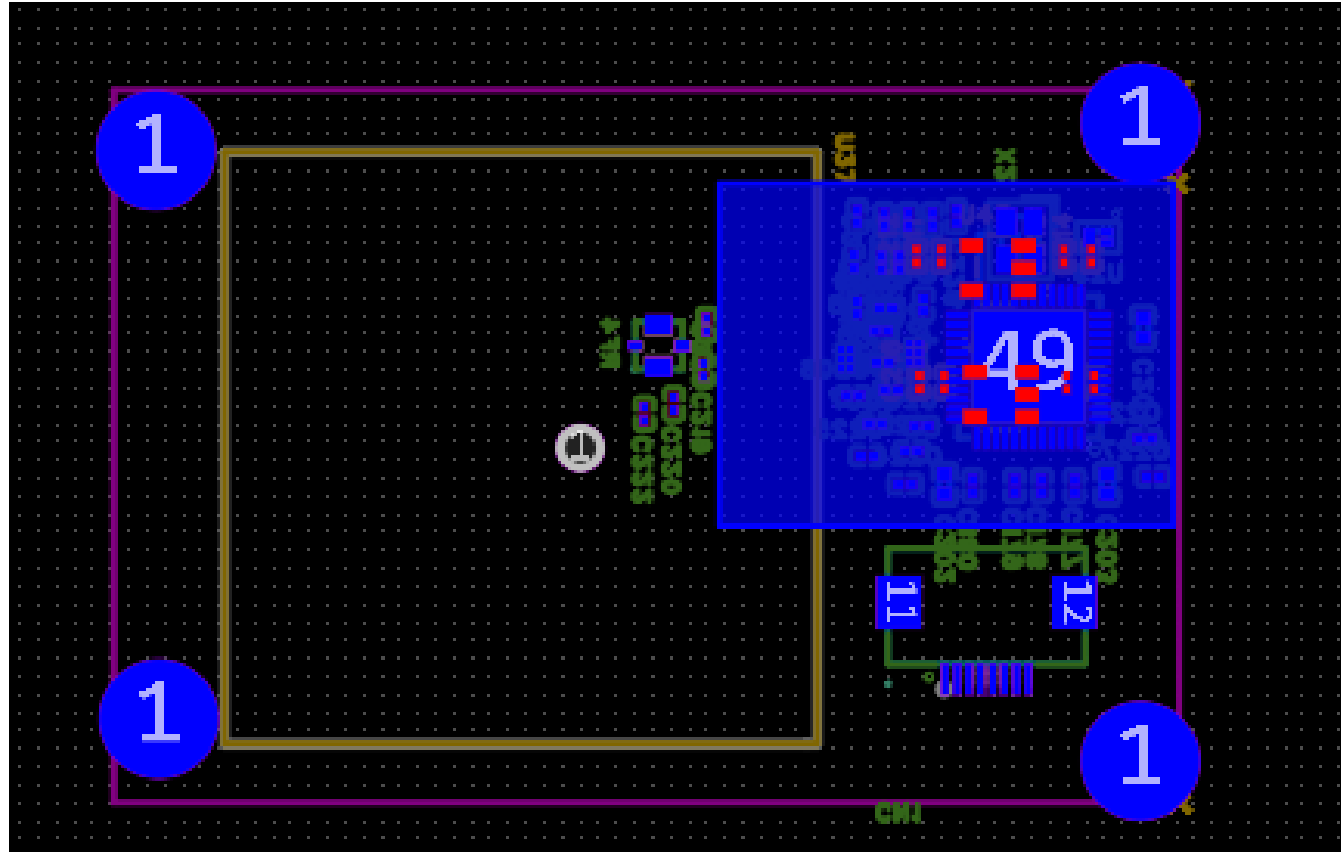


Overall design — Component Placement Evaluation—V1.8 update

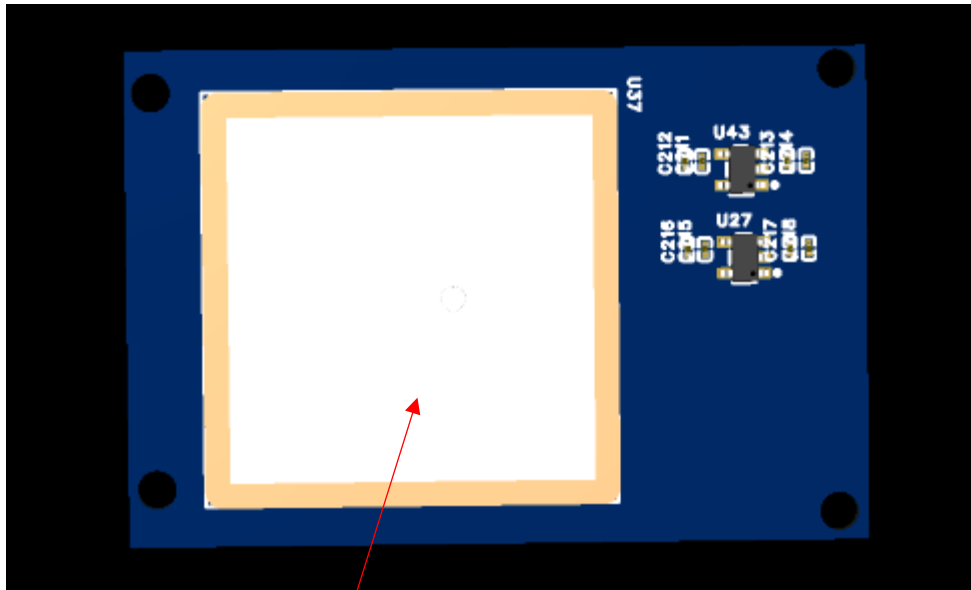


Overall design – Component Placement Evaluation—V1.8 update

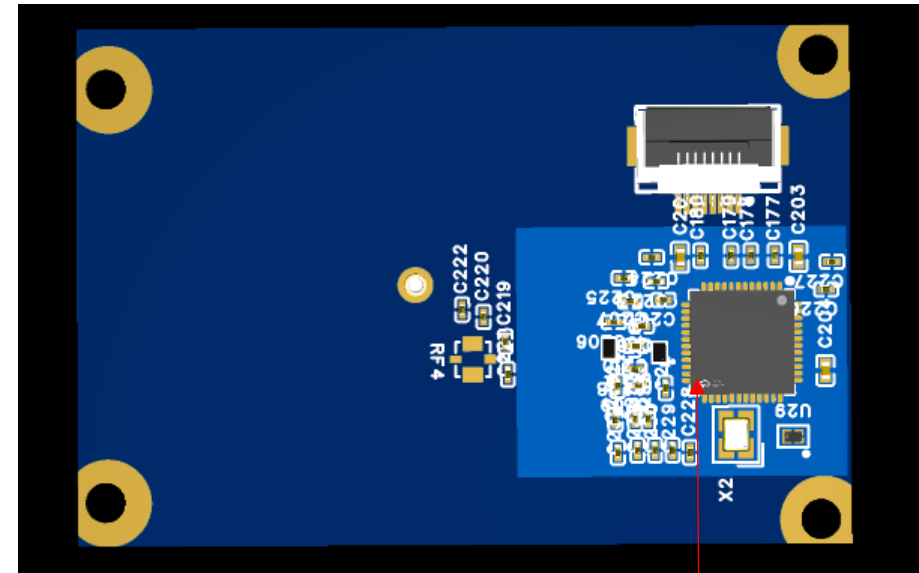
Estimated GNSS PCBA dimensions: 30×45×1.6 mm, 4-layer through-hole PCB.



Overall design — Component Placement Evaluation—V1.8 update



GNSS ANT



GNSS IC

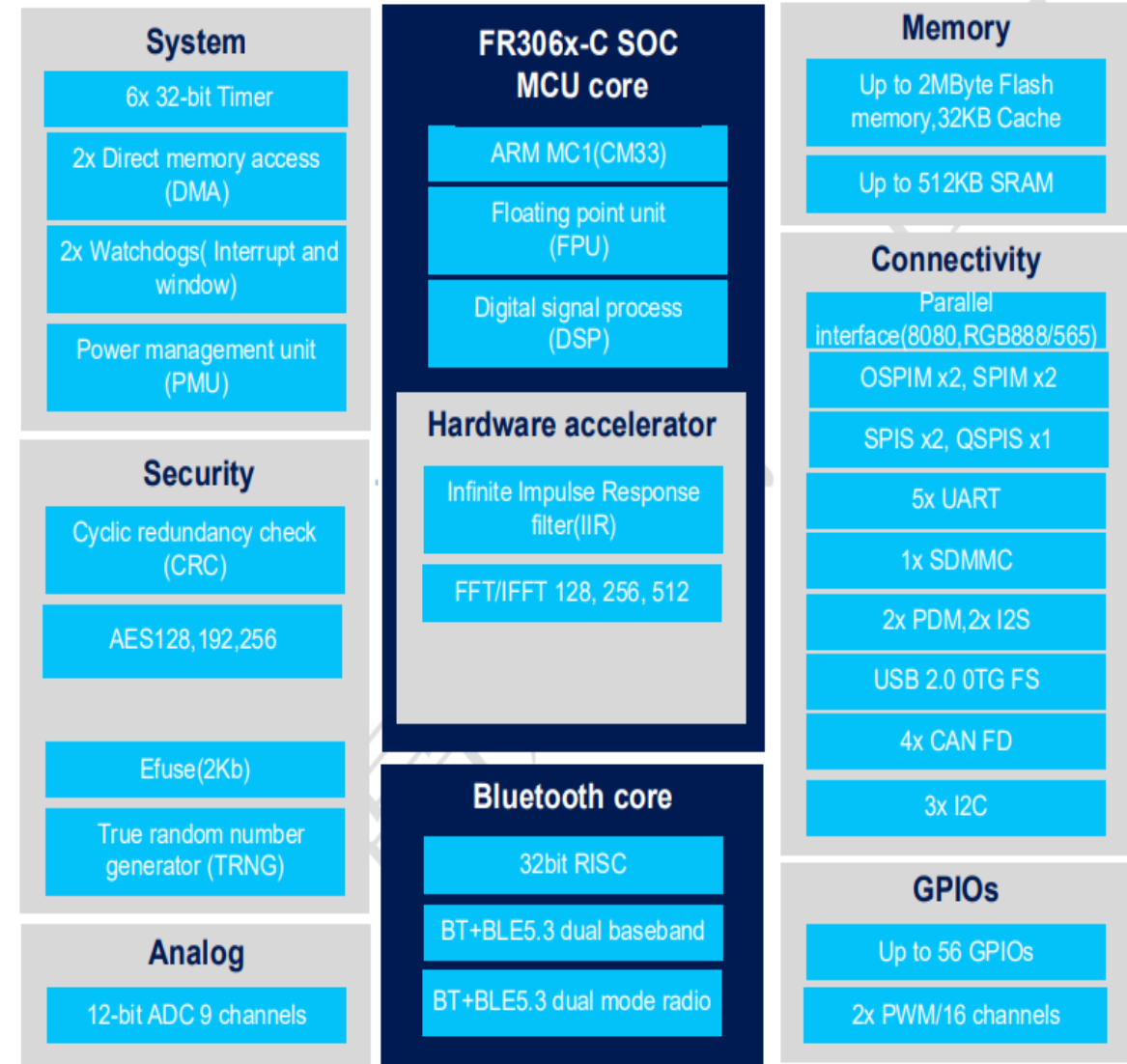
Key component selection – MCU selection (FR3068E-C)

Overview

- The FR306x series is a high-performance, low-power, and highly secure wireless MCU featuring an integrated Bluetooth BR/EDR/BLE transceiver and controller, as well as a CAN FD bus controller, making it ideal for applications in industrial and automotive electronics. It complies with the AEC-Q100 Grade 2 automotive qualification standard.
- The FR306x supports Bluetooth V5.3, enabling BR 1 Mbps GFSK, EDR 2 Mbps $\pi/4$ -DQPSK, and 3 Mbps 8DPSK. BLE1M/2Mbps GFSK, multiple modes including 125K/500K, supports independent enabling and disabling of different modes, and enables Bluetooth multi-master, multi-slave, and multi-connection. The FR306x supports the AUTOSAR software framework; it integrates up to four independent CAN FD controllers, compatible with CAN2.0 A/B.

Features

- Dual-core processor architecture - MC1 (CM33) core with a maximum clock frequency of 156 MHz, supporting FPU and DSP instruction sets, serving as the main application processor -
- 32-bit RISC core operating at 48 MHz, equipped with built-in independent SRAM, running the Bluetooth protocol stack as a co-processor for Bluetooth protocol processing Flash capacity: 1MB
- SRAM: 512KB
- Built-in maximum 2MB flash for storing user programs, with 32KB cache



Key component selection – Module selection (EG800Q-EU)

EG800Q series specifications

17.7 × 15.8 × 2.4 mm
10 Mbps DL/ 5 Mbps UL

Variant		EG800Q-EU	EG800Q-NA	EG800Q-IN	EG800Q-GL
Region		Europe	North America	India	Global
Chipset		QCX216	QCX216	QCX216	QCX216
LTE	FDD band	B1/ 3/ 5 /7/ 8/ 20/ 28	B2/ 4/ 5/ 12/ 13/ 66	B1/ 3/ 5/ 8	B1/ 2/ 3/ 4/ 5/ 7/ 8/ 12/ 13/ 14/ 17/ 18/ 19/ 20/ 25/ 26/ 28/ 66/ 71
	TDD band	-	-	B40/ 41	B34/ 38/ 39/ 40/ 41
UMTS	WCDMA band	-	-	-	-
	TD-SCDMA band	-	-	-	-
GSM		-	-	-	-
Embedded GNSS		-	-	-	-
VoLTE		-	-	-	-
Wi-Fi Scan		Supported	Supported	Supported	Supported
Bluetooth		-	-	-	-
DFOTA		Supported	Supported	Supported	Supported
Open solution		-	-	-	-
Certification	Carrier	Deutsche Telekom/ Spark	AT&T/ T-Mobile/ Verizon	-	T-Mobile*/ KDDI*/ Telstra*/ Rogers*
	Regulatory/ Conformance	CE/ UKCA/ RCM/ Anatel	FCC/ IC/ GCF/ PTCRB	-	GCF*/ CE*/ PTCRB*/ FCC*/ JATE*/ TELEC*/ RCM*
	Others	WHQL	WHQL	WHQL	WHQL
Project stage		MP	MP	CS	Kick-off

* : In progress

[Back](#)

[Return to contents](#)

Build a smarter world

Key component selection – GNSS selection (AG3352Q)

Multi –GNSS receiver

- Multi-GNSS GPS/GLONASS/Galileo/ BeiDou receiver
- Support for multi-GNSS including QZSS and SBAS ranging
- Integrated 12 multi-tone active interference cancellers
- Indoor and outdoor multi-path detection and compensation
- Maximum update rate 10Hz

Software features

- EPO™ orbit prediction
- EASY™ self-generated orbit prediction
- LOCUS™ logger function

Microcontroller subsystem

- ARM® Cortex®-M4 with FPU and MPU with maximum frequency 432MHz
- 13 DMA channels
- One RTC timer
- Nine 32-bit general purpose timers
- Two 32-bit OS timers
- Development support: SWD, JTAG
- True random number generator

Communication interfaces

- I2C master (3.4Mbps) and I2C slave interfaces
- Three UART interfaces (3Mbps, with hardware flow control)
- SPI slave

Power management

- Integrated LDO regulators with low quiescent current for RTC, RF frontend and GPIOs
- Operating temperature from -40°C to 85°C
- Thermal protection sensor

Clock source

- Support 26MHz or 48MHz TCXO
- Support 32kHz crystal oscillator

Package

- AG3352Q: 4.4mm x 4.0mm x 0.8mm QFN with 0.35 pitch
- AG3352B: 3.3mm x 3.1mm x 0.807mm BGA with 0.35 ball pitch

1.2 System architecture

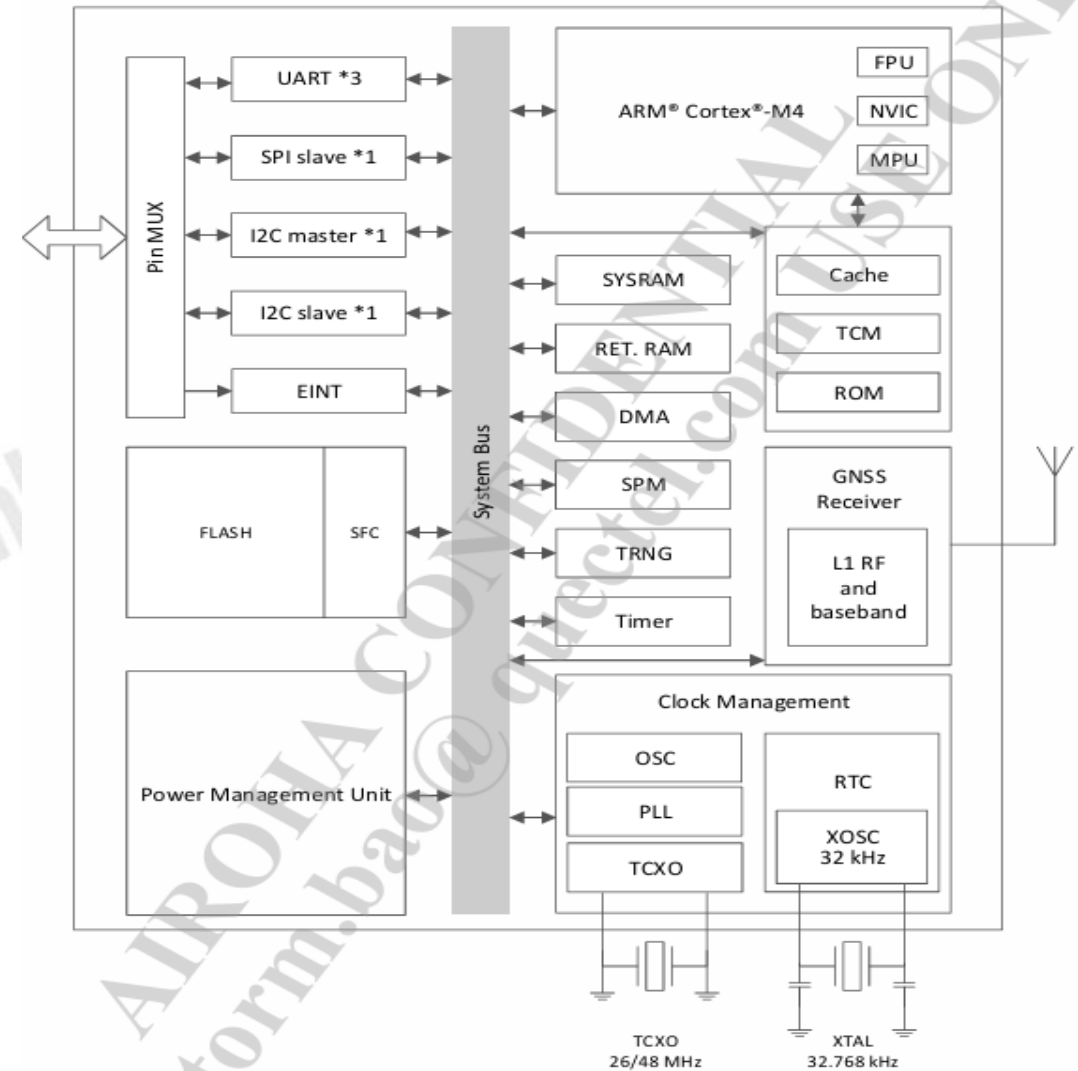


Figure 1.2-1. Chip architecture

Key component selection – RFID selection (PN7160)

Features

- Highly integrated demodulator and decoder
- Buffered output drivers to connect an antenna with a minimum number of external components
- Integrated RF level detector
- Integrated Polling Loop for automatic device discovery
- RF protocols supported
 - ISO/IEC 14443A, ISO/IEC 14443B PICC mode
 - ISO/IEC 14443A, ISO/IEC 14443B PCD mode designed according to NFC Forum digital protocol T4T platform and ISO-DEP (see ref.[1])
 - FeliCa PCD mode
 - MIFARE PCD encryption mechanism (MIFARE 1K/4K)
 - NFC Forum tags T1T, T2T, T3T, T4T, and T5T (see ref.[1])
 - NFCIP-1, NFCIP-2 protocol (see ref.[9]and ref.[11])
 - Only up to Firmware version 12.50.0A: NFC Forum certification for P2P, reader and card mode (see ref.[1]– ISO/IEC 15693/ICODE VCD mode (see ref.[10])
 - NFC Forum-compliant embedded T4T for NDEF short record

9 Block diagram

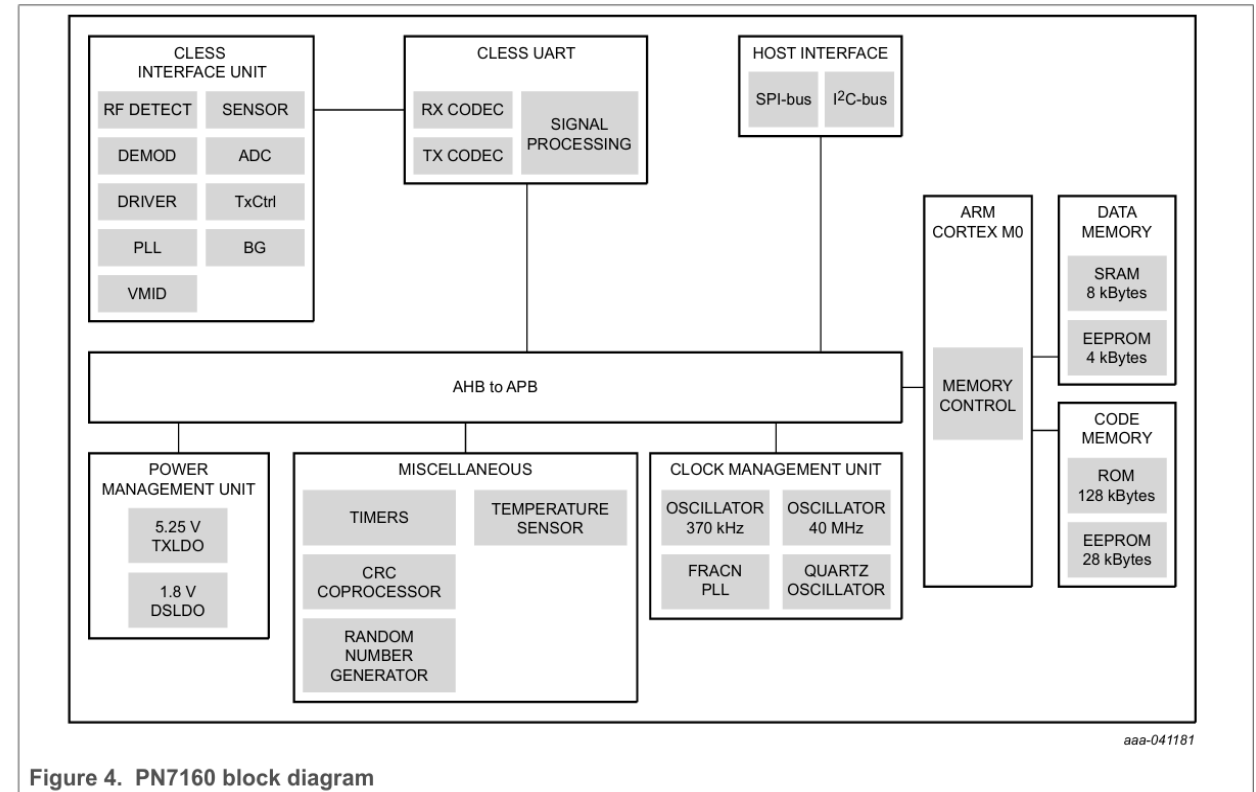


Figure 4. PN7160 block diagram

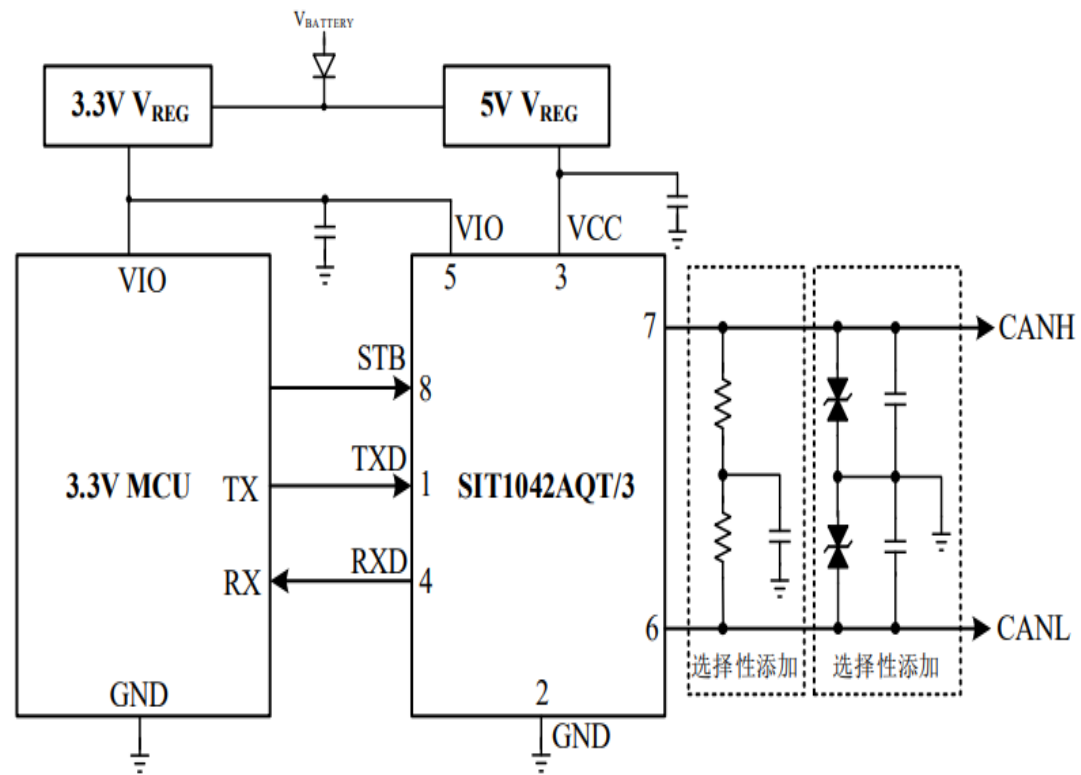
20 References

- [1] NFC Forum Device Requirements V2.0
- [2] NFC Controller Interface (NCI) Technical Specification V2.0
- [3] ISO/IEC 14443 parts 2: 2001 COR 1 2007 (01/11/2007), part 3: 2001 COR 1 2006 (01/09/2006) and part 4: 2nd edition 2008 (15/07/2008)
- [4] I²C Specification, UM10204 rev4 (13/02/2012)
- [5] SPI Motorola de-facto standard described in Motorola 68HC11 data sheet
- [6] UM11577 PN7161 NFC controller user manual
- [7] AN12988 PN7160 hardware design guide
- [8] AN13219 PN7160 antenna design and matching guide
- [9] ISO/IEC 18092 (NFCIP-1) edition, 15/03/2013. This is similar to Ecma 340.
- [10] ISO/IEC 15693 part 2: 2nd edition (15/12/2006), part 3: 1st edition (01/04/2001)
- [11] ISO/IEC 21481 (NFCIP-2) edition, 01/07/2012. This is similar to Ecma 352.
- [12] ETSI HCl TS 102 622; UICC - Contactless Front-end (CLF) Interface; Host Controller Interface (HCI) (Release 12)
- [13] Apple Enhanced Contactless Polling Specification: Version 1.1.

Overall design – Detail view (CAN)

Description

- Fully compliant with ISO 11898 standard
- Built-in over-temperature protection function
- $\pm 70\text{V}$ withstand voltage on bus pins
- Dominant timeout function on driver (TXD)
- Dominant timeout function on standby bus (BUS)
- Low-power standby mode with wake-up function
- SIT1042AT/3 I/O voltage range supports both 3.3V and 5V MCU
- Under-voltage protection on VCC and VIO power supply pins
- High-speed CAN, supports 5Mbps CAN FD (Flexible Data Rate)
- Typical loop delay from TXD to RXD is less than 100ns
- High electromagnetic immunity (EMI)
- Unpowered nodes do not disturb the bus
- Supports DFN3 $\times$ 3-8 (small outline, leadless package)



Overall design – Detail view (DC-DC)

Features

- 4.3V to 60V Input Range
- 3A Continuous Output Current
- Ultra-Low 50 μ A Operating Quiescent Current
- 140m Ω High-side MOSFET
- Minimum Switching-On Time: 100ns
- Current Mode Control
- SGM61630A: Soft-Start Version
- SGM61630B: Power-Good Version
- Adjustable Switching Frequency from 200kHz to 2500kHz
- Frequency Synchronization to External Clock
- Easy-to-Use Internal Compensation
- Support High Duty Cycle Operation
- Precision Enable Input
- 2 μ A Shutdown Current
- Thermal, Over-Voltage and Short Protection
- Available in a Green SOIC-8 (Exposed Pad) Package

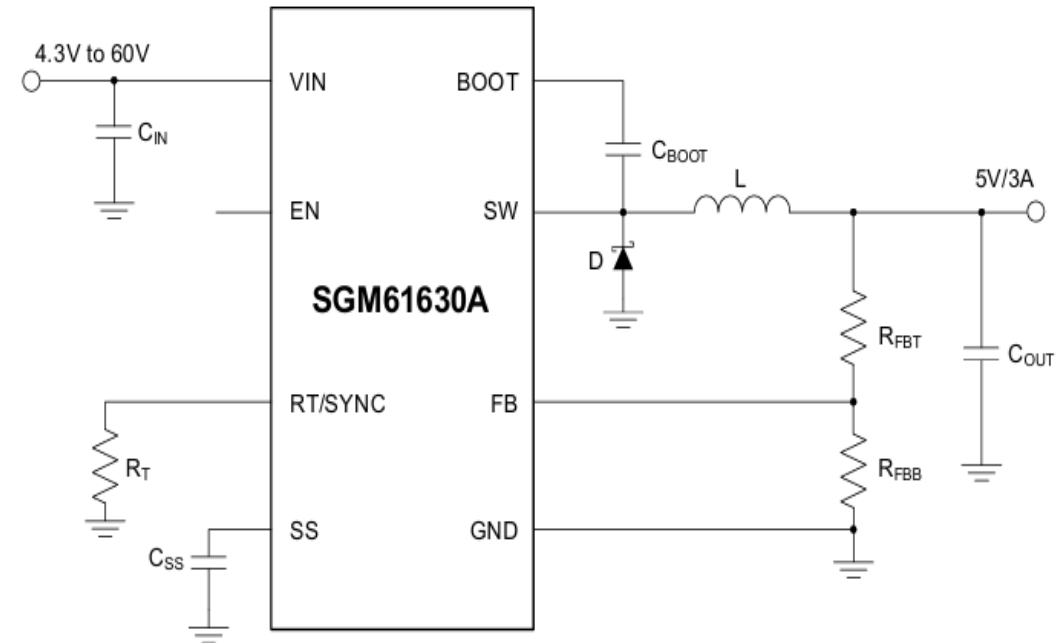
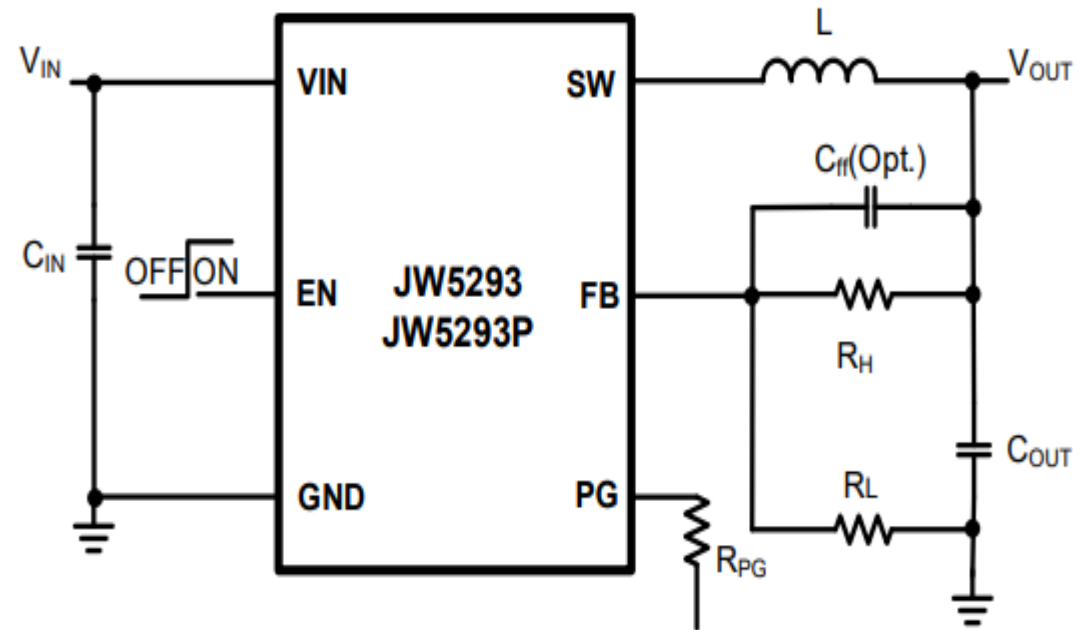


Figure 1. SGM61630A Typical Application

Overall design – Detail view (DC-DC)

Features

- 2.7V to 6.0V Operating Input Range
- Up to 3A Output Current
- Up to 95% Peak Efficiency
- Very Low R_{dson}
- High Efficiency at Light Load
- 1% FB Accuracy
- Internal Soft-Start
- 1.5MHz Switching Frequency
- Input Under Voltage Lockout
- Optional Power Good (PG) Function
- Output Discharge
- Pre-bias Soft Start
- Current Run-Away Protection
- Short Circuit Protection
- Thermal Protection
- Available in SOT563 Package



Overall design – Detail view (LDO)

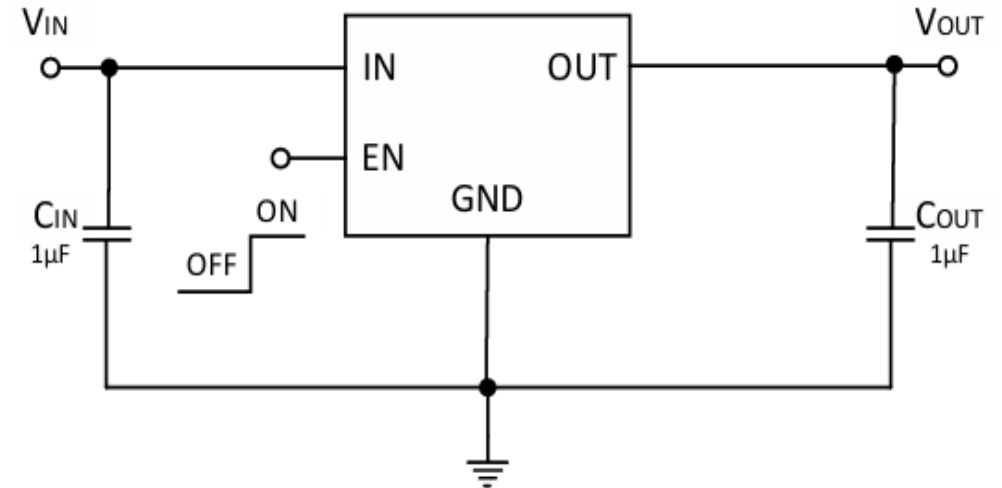
General Description

The WR0338 series can provide 300mA output current and the maximum input voltage is 18V, the series is a high accuracy, low noise, high speed, low dropout CMOS Linear regulator with high ripple rejection. The devices offer a new level of cost-effective performance in the battery-powered equipment, such as RF applications and other systems requiring a quiet input voltage.

WR0338 has the fold-back maximum short output current which depends on the output voltage. So WR0338 features short circuit protection and output current limiter.

Features

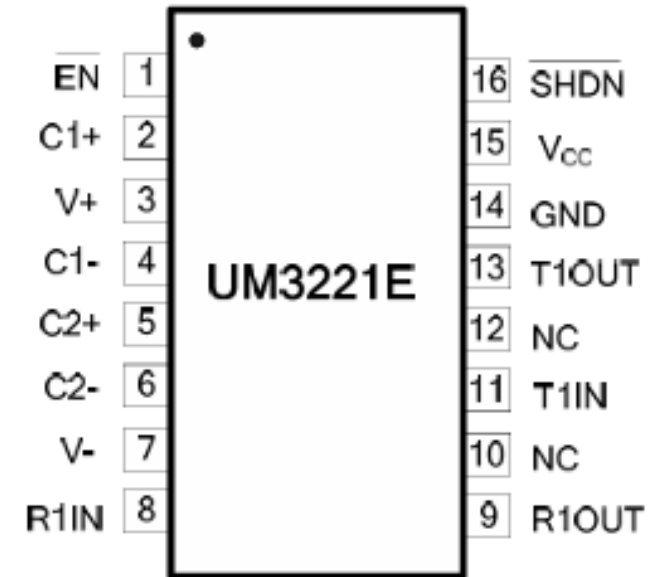
- Input Voltage: 2.5V-18V
- Output Voltage: 1.2V-5V
- Output Current: 300mA
- PSRR: 75dB@1kHz
- Dropout Voltage: 170mV @ I_{our} = 100mA
- Shutdown Current: < 1uA
- Quiescent Current: 3.0uA Typ.
- Recommend Capacitor: 1uF



Overall design – Detail view (RS232)

Features

- Meets True EIA/TIA-232-F Standards from a +3.0V to +5.5V Power Supply
- Meets EIA/TIA-562 Levels of $\pm 3.7V$ with Supply Voltages as Low as 2.7V
- Enhanced ESD Specifications:
 - $\pm 15kV$ Human Body Mode
 - $\pm 15kV$ IEC61000-4-2 Air Discharge Mode
 - $\pm 8kV$ IEC61000-4-2 Contact Discharge Mode
- 1 μA Low Power Shutdown (UM3221E/UM3222E)
- 250kbps Minimum Transmission Rate
- Guaranteed 30V/ μs Max Slew Rate
- Latch-Up Performance Exceeds 200mA
- Hot Swap and Fail-Safe
- I/O Logic Compatible with 1.8V Logic @ VCC=3.3V



SSOP16/TSSOP16

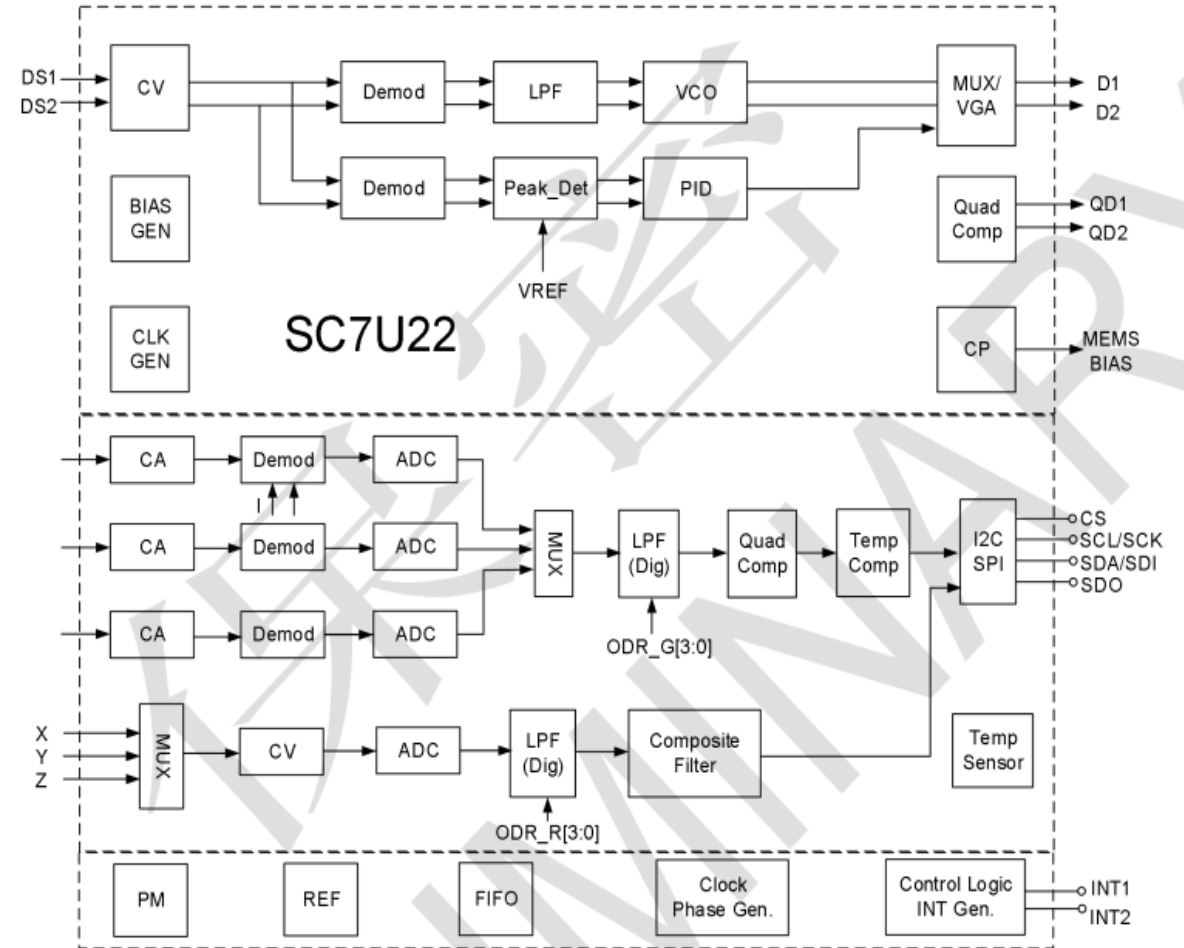
Overall design – Detail view (G+Sensor)

Description

SC7U22 is a highly integrated, low-power inertial measurement unit (IMU) featuring a high-performance 3-axis accelerometer and 3-axis gyroscope.

The accelerometer offers selectable full-scale ranges of $\pm 2g$, $\pm 4g$, $\pm 8g$, and $\pm 16g$, while the gyroscope provides angular rate ranges of $\pm 125dps$, $\pm 250dps$, $\pm 500dps$, $\pm 1000dps$, and $\pm 2000dps$, allowing users to flexibly measure external acceleration and angular velocity.

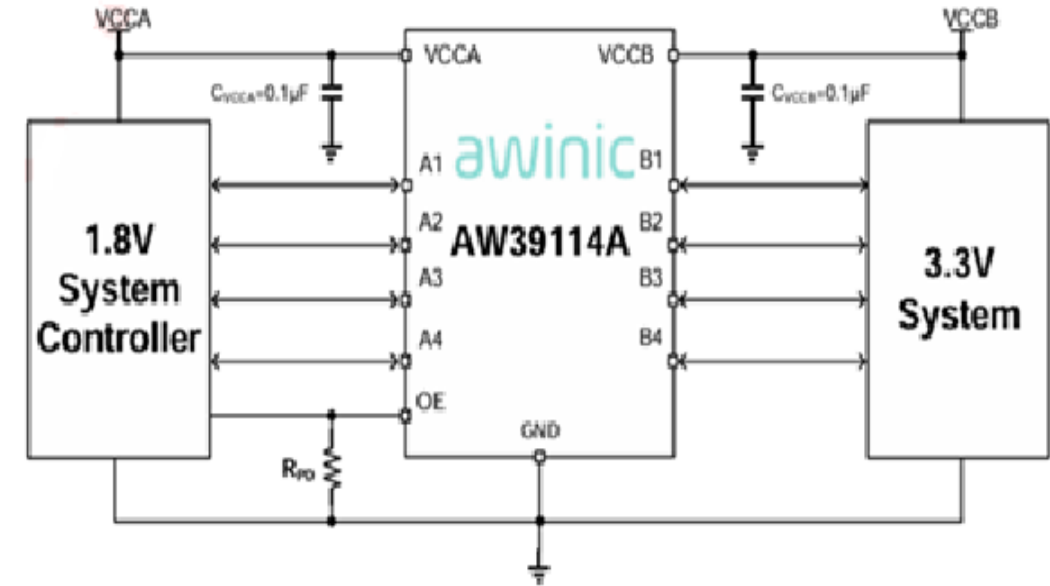
The output data rate for the accelerometer is selectable from 0.78Hz to 1.6kHz, and for the gyroscope from 25Hz to 3.2kHz. The device operates over a temperature range of $-40^{\circ}C$ to $+85^{\circ}C$.



Overall design — Detail view (LEVEL SHIFT)

Features

- Voltage Level Translator Without Direction-Control Signal
- Maximum Data Rates
24Mbps (Push Pull)
2Mbps (Open Drain)
- Power Supply Range:
A Port and VCCA: 1.1V to 3.6V
B Port and VCCB: 1.65 V to 5.5 V
 $VCCA \leq VCCB$
- Pull Up Resistors are Integrated in A Port and B Port
- No Power-Supply Sequencing Required: Either VCCA or VCCB Can be Ramped First
- Support Ultra-Low Power Consumption Mode with OE Pin is Low Voltage Level
- Latch-Up Performance Exceeds +200mA Under JESD 78 Standard
- WBQFN 1.8mmx1.8mmx0.55mm -12L Package

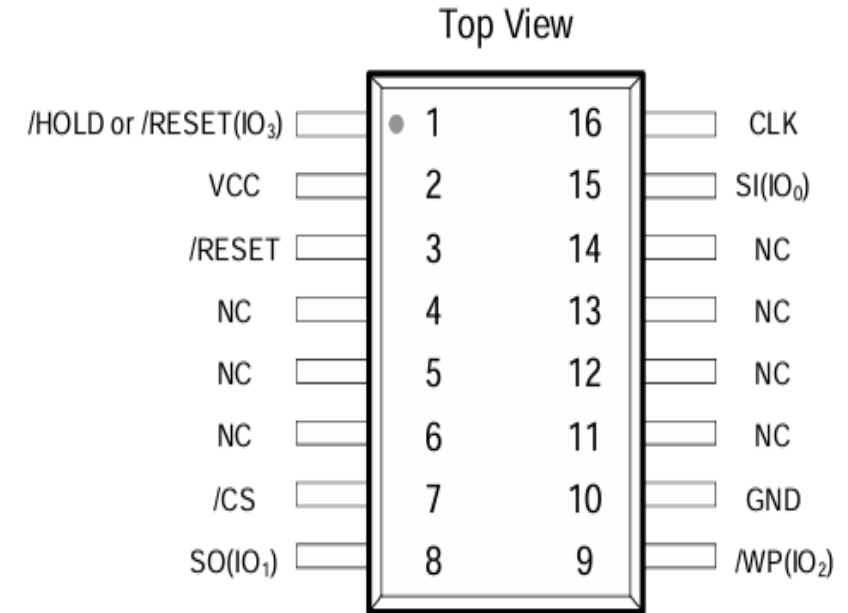


Overall design – Detail view (FLASH)

Description

The XM25QH256D (256M-bit) is a serial interface Flash memory device designed for operating on a single power supply from 2.3-3.6V with current consumption as low as 0.2 μ A at deep power-down in which program code is shadowed from Flash memory into embedded or external RAM for execution.

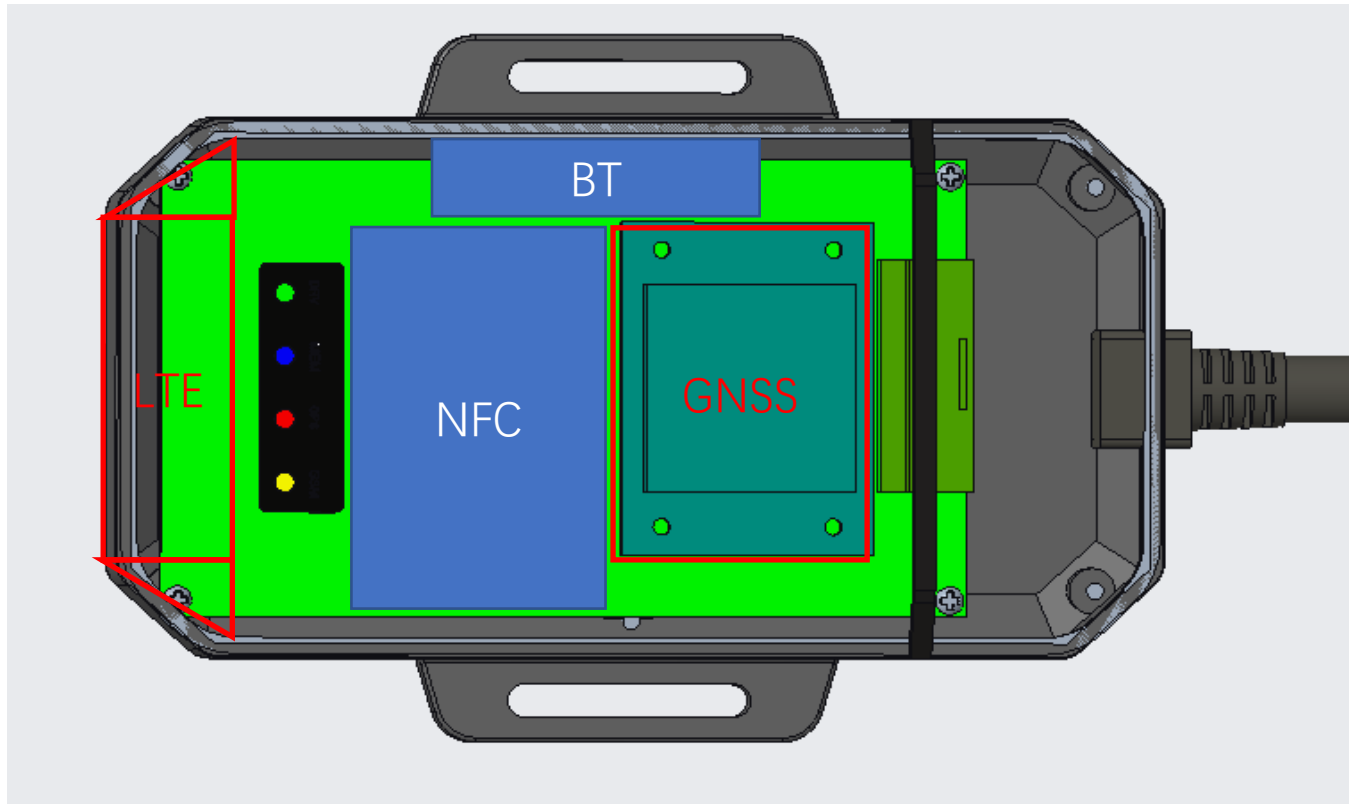
The flexible erase architecture of the device with its page erase granularity is ideal for data storage as well, eliminating the need for additional data storage devices.



Selection of key components—V1.8 update

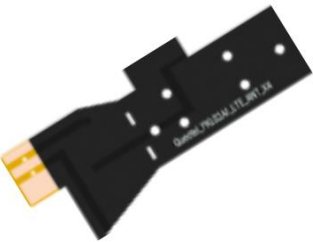
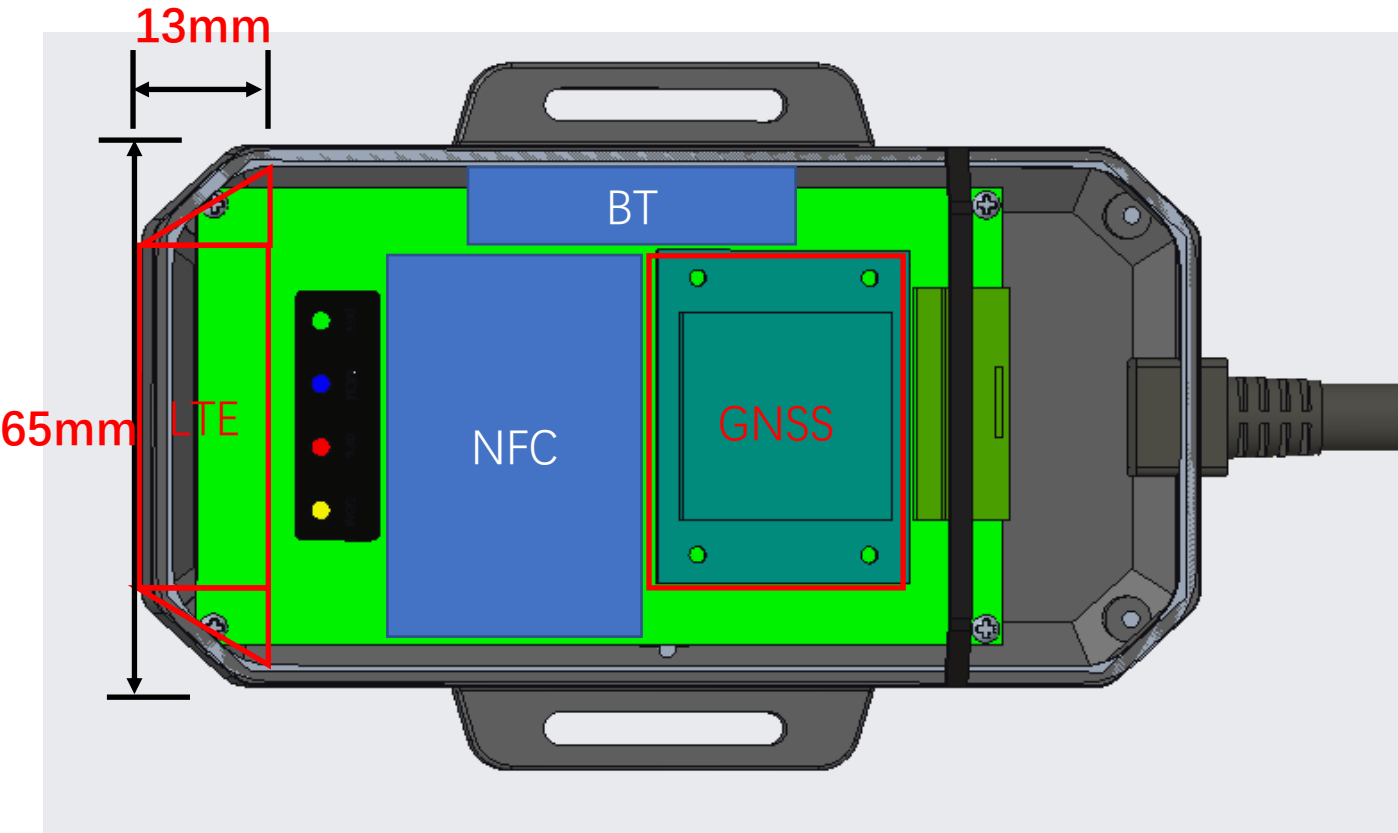
No.	Component	Description	Quantity
1	MCU	FR3068E-C	1
2	LTE Module	EG800Q-EU	1
3	DCDC	SGM61630	2
4	LDO	PMIC LDO 2.5-18V 3.3V 300mA 3uA SOT23-5 PINS5 2.8x2.9mm H1.5mm RO	2
5	RS232	UM3221E	3
6	G + Sensor	SC7U22	1
7	Level Shift	AW39114A	1
8	FLASH	XM25QH256D	1
9	GNSS	AG3352Q	1
10	DCDC	JW5293	1
11	CAN Transceiver	SIT1042AT/3	1
12	RFID	PN7160	1

Antenna Placement —V1.8 update



As shown in the left figure, the antenna distribution positions are indicated. The GNSS uses a ceramic antenna. The NFC antenna is attached to the upper shell via a pin. The LTE antenna and BT antenna are attached to the lower shell via spring clips and are then adjusted along the side of the housing. The current performance is for 3D drawing evaluation. The final performance will be determined based on actual debugging.

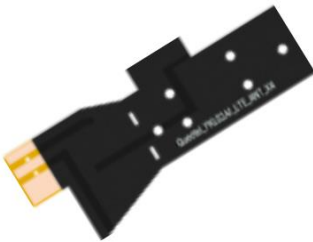
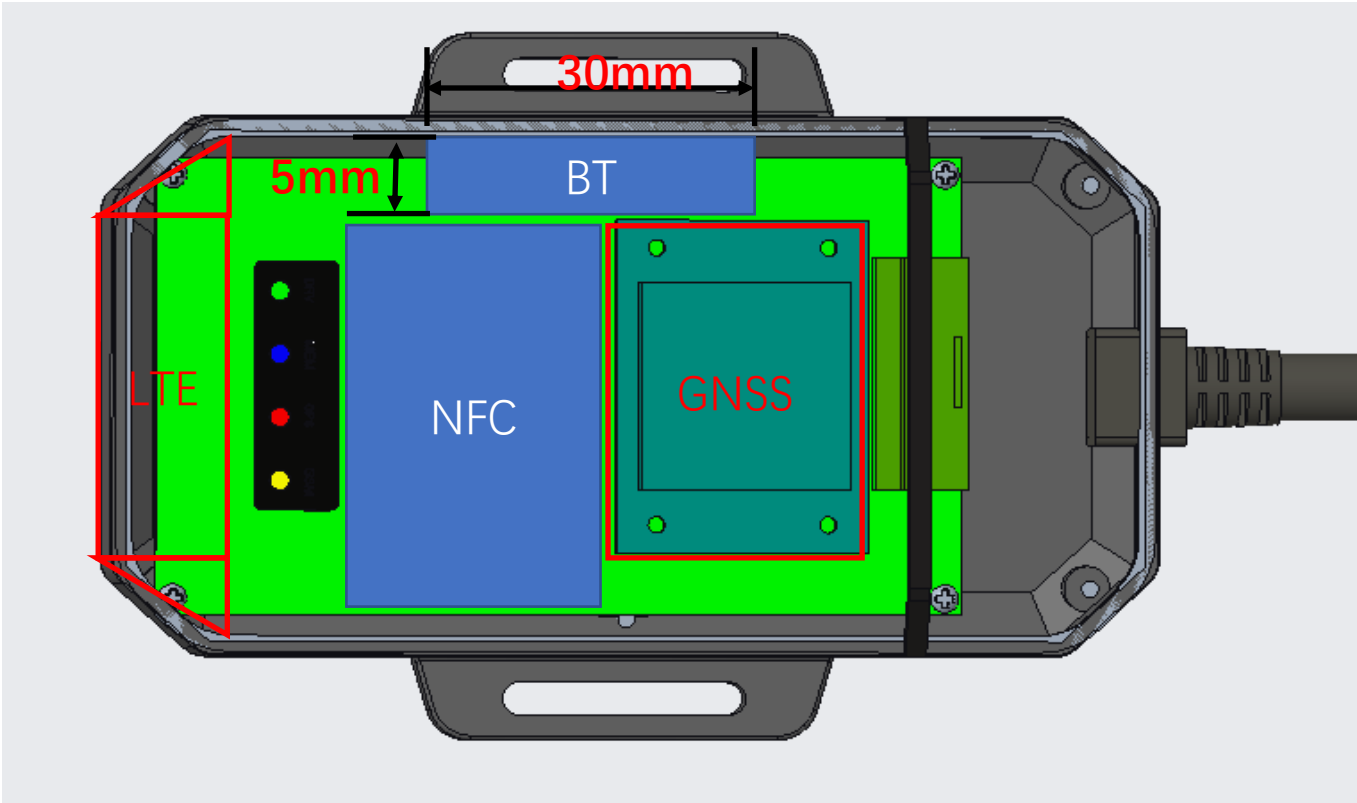
Antenna Placement —V1.8 update



As shown in the left figure, the LTE antenna debugging is located on the left side of the device, and the clearance size of the main board is 64*13mm.

LTE	
Frequency	Efficiency
600-960MHz	20-40%
1700-2700MHz	45-55%

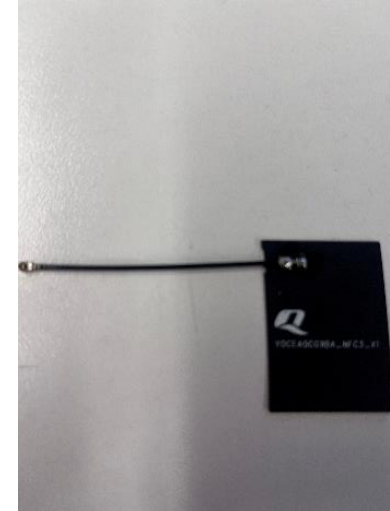
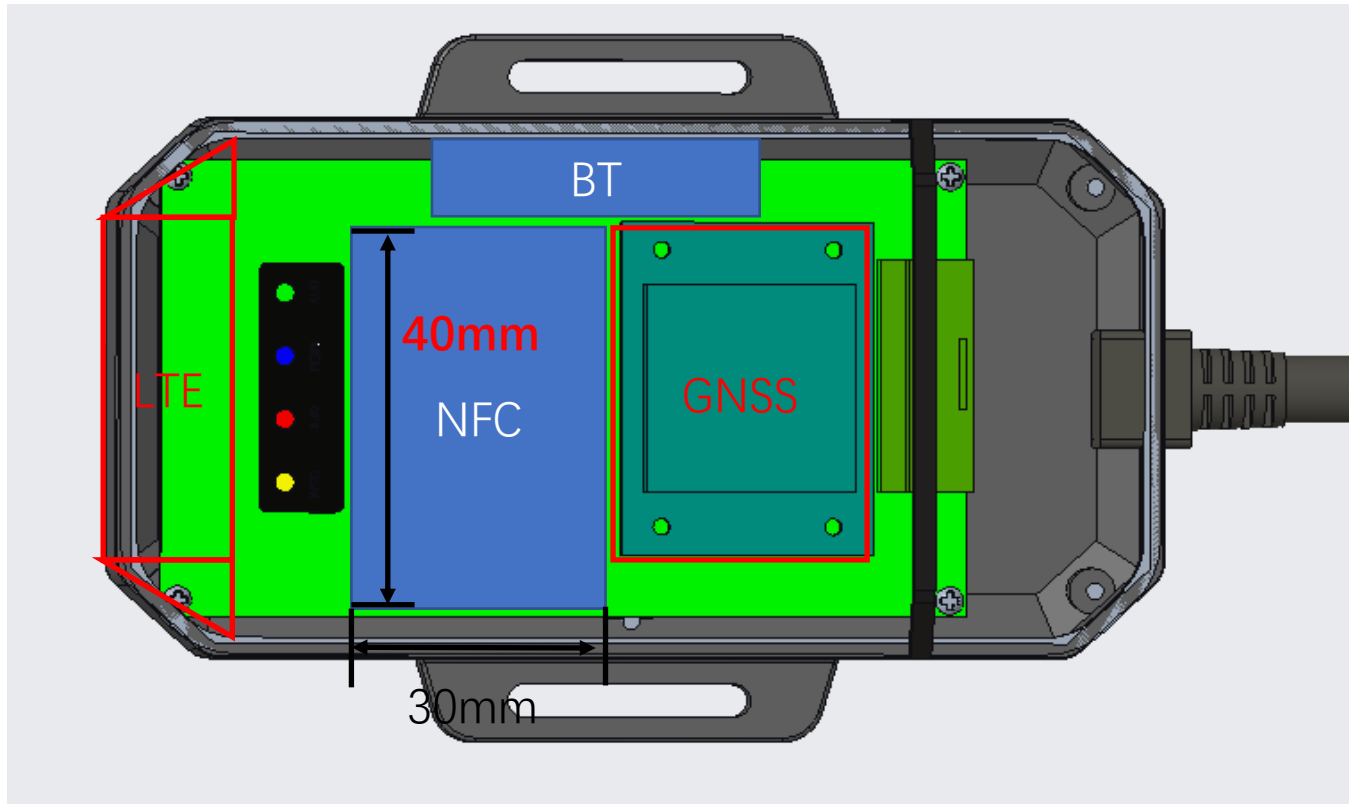
Antenna Placement —V1.8 update



As shown in the left figure, the BT antenna is being debugged at the lower edge of the device, and the clearance size of the main board is 30*5mm.

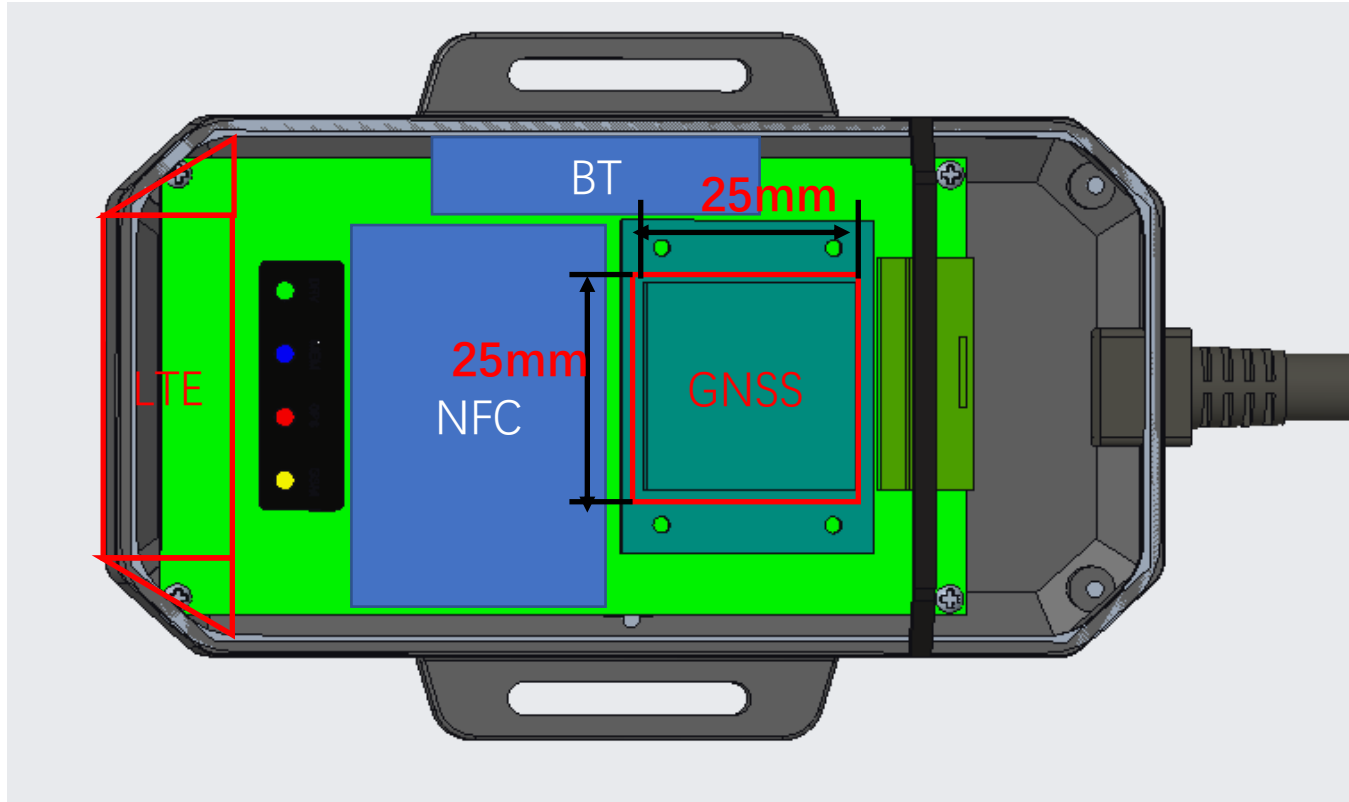
BT	
Frequency	Efficiency
2400-2500MHz	40-50%

Antenna Placement —V1.8 update



As shown in the left figure, the NFC antenna debugging is carried out at the center top shell position of the device. The size of the antenna is 40*30mm.

Antenna Placement —V1.8 update



As shown in the left figure, the GNSS antenna is being debugged at the middle mainboard position of the device, and the antenna size is 25*25mm.

ID Reference—V1.8 Update



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CMF--V1.8 Update



1.主壳体

C: Pantone black 7C

M: plastic

F: 素材注塑, VDI24#

LED灯光区域和光带处模具抛光

LOGO字符 丝印pantone冷灰7C

Icon字符 丝印pantone 冷灰7C

2.底壳

C: Pantone black 7C

M: plastic

F: 素材注塑, VDI24#

3.线束

C: Pantone black 7C

M: plastic

F: 注塑线束

线束粗细以实际样品为准

4.螺丝*8

C: Pantone black 7C

M: 金属

F: 标准件

5.铭牌

C: Pantone black 7C 背景黑色 字体白色

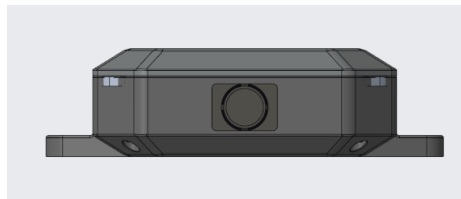
M: PET

F: 印刷

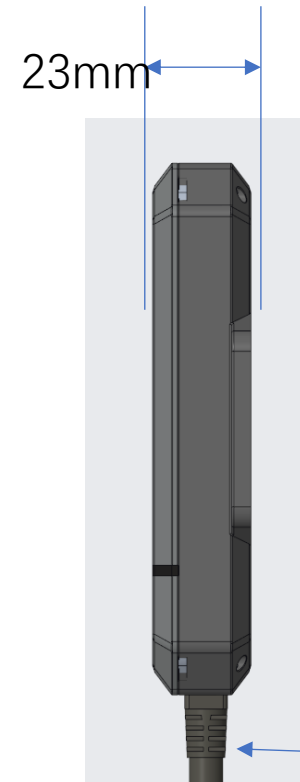
Estimated Dimensions--V1.8 Update



Buzzer hole
Exact position may need adjustment based on detailed design



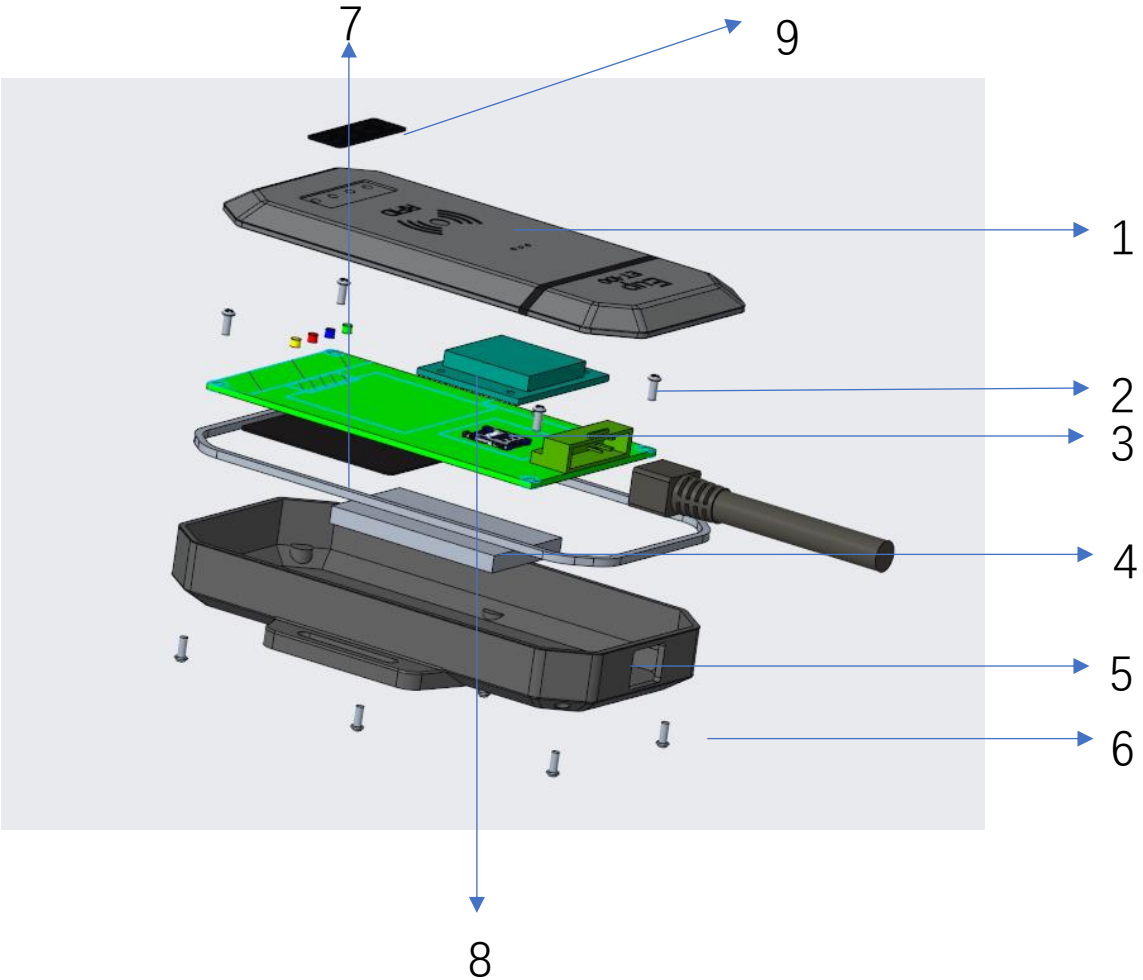
This is a glossy surface, not a separate plastic part.



The wiring harness is based on the final detailed design.

Remark: Subject to the actual design.

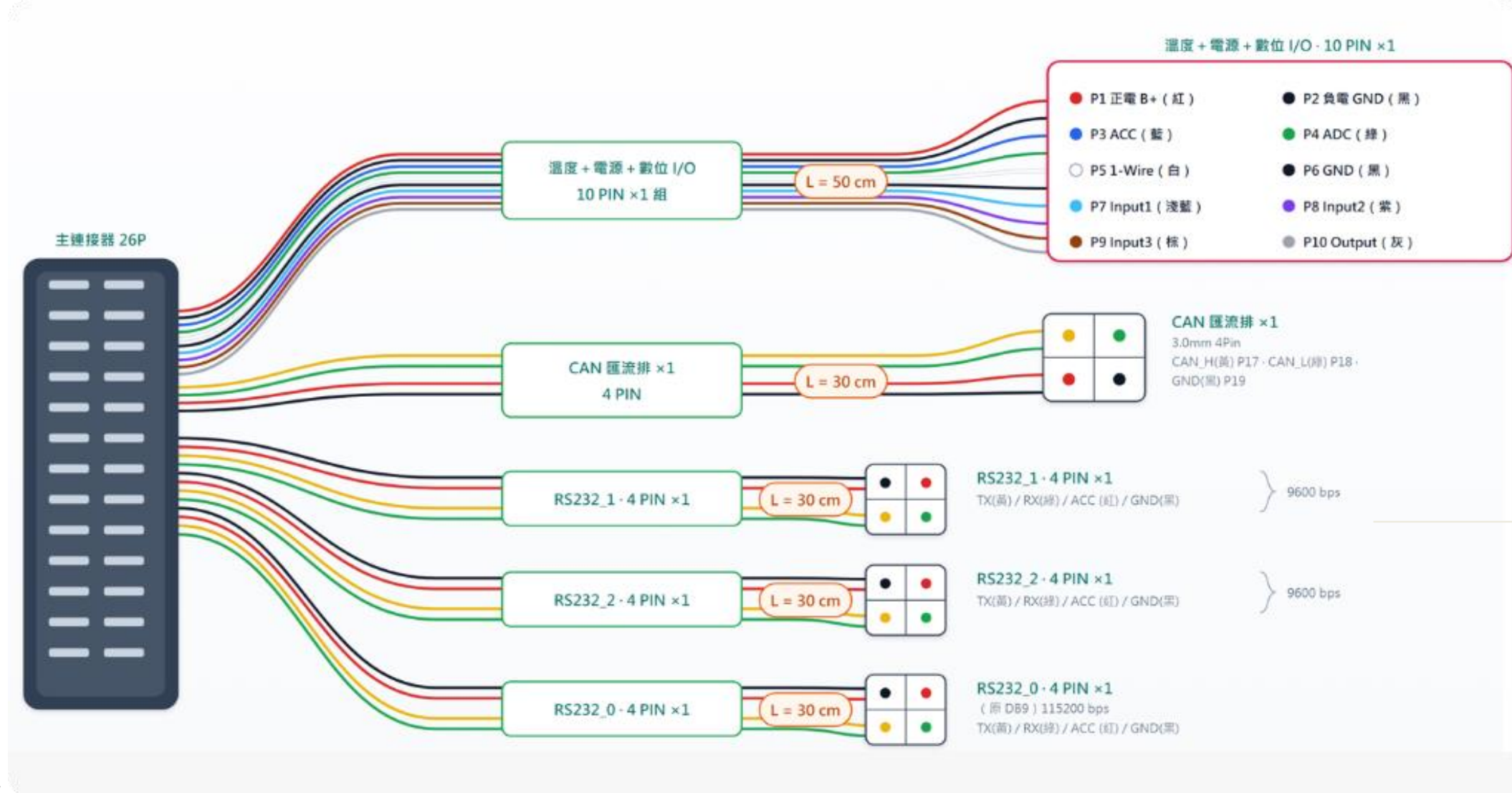
Product stacking--V1.8 Update



No.	Name	Qty.
1	Top shell	1
2	PCBA screws	4
3	PCBA	1
4	Battery	1
5	Bottom shell	1
6	Shell screws	4
7	Waterproof silicone	1
8	GNSS PCBA	1
9	Light film	1

Note: The positions and quantities of the screws shall be in accordance with the detailed design.

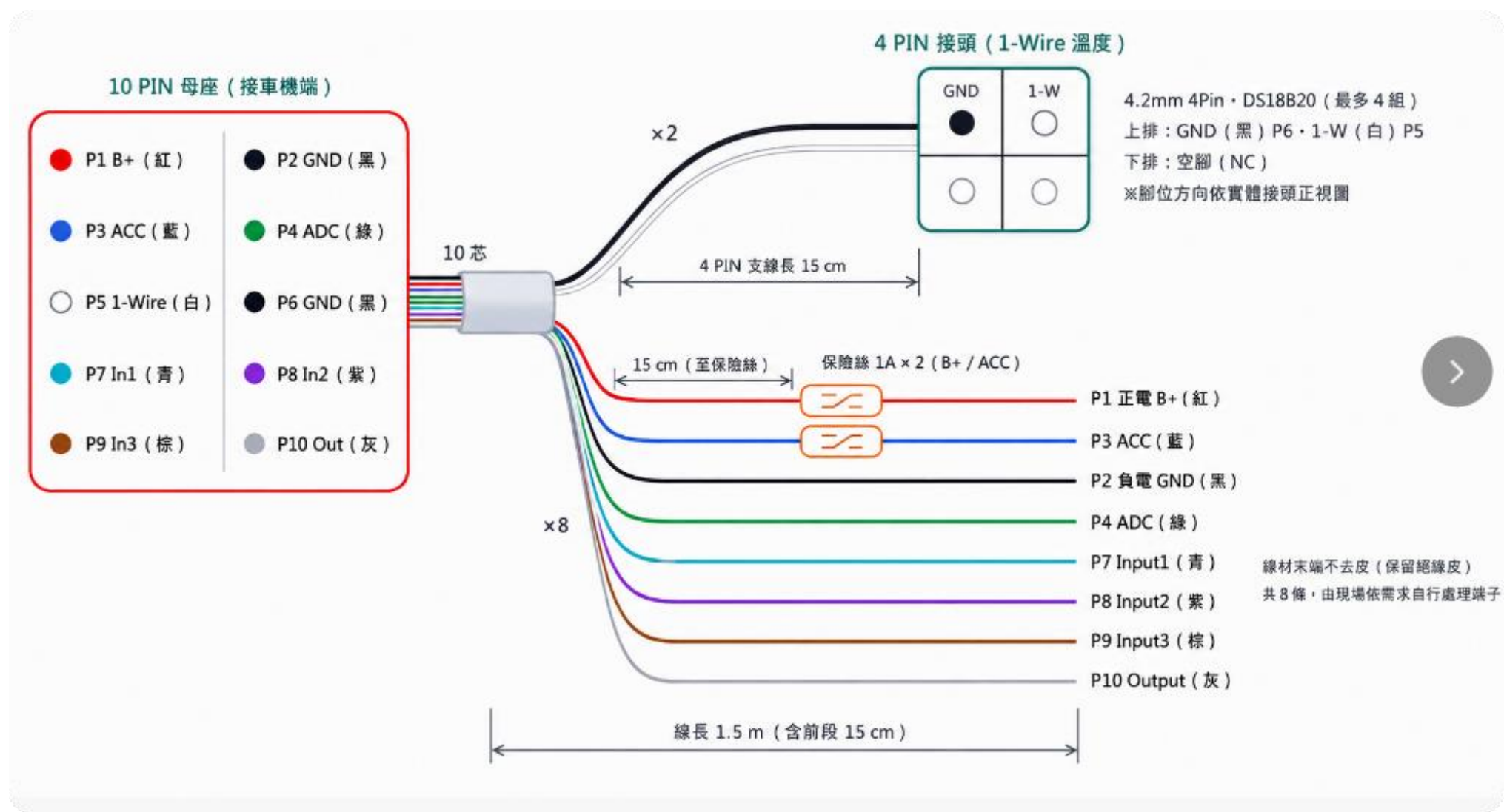
Wiring Harness– 26 pin adapter cable--V1.8 Update



Remark:

- 1.For the dedicated RS232 interfaces, the corresponding labels (RS232_0, RS232_1, and RS232_2) will be marked on the cable or interface.
- 2.The above 26 pin wiring harness is standard configuration.
- 3.The socket need use wiring harness design

Wiring Harness-- 10pin adapter cable



Remark: The above 10 pin wiring harness is standard configuration.

Wiring Harness– 4pin to DB9 cable

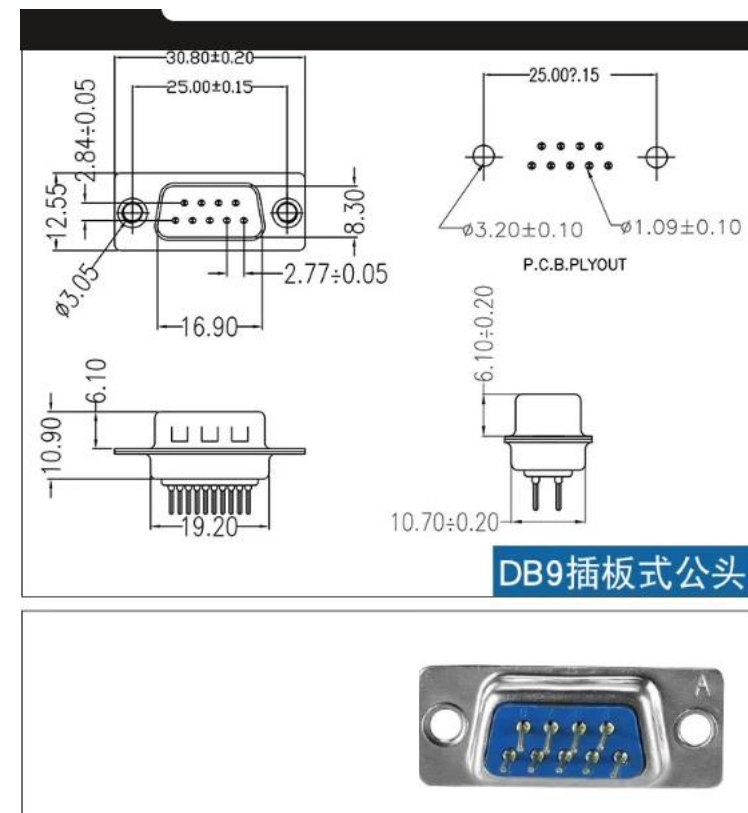


Connect the female connector of 5557 4-pin

Connect DB9 male head as the right picture

Four 24 awg cable (2A current), 150 mm,
and outsourcing black PVC soft leather (normal process)

Remark: The above 4 pin wiring harness is optional configuration.



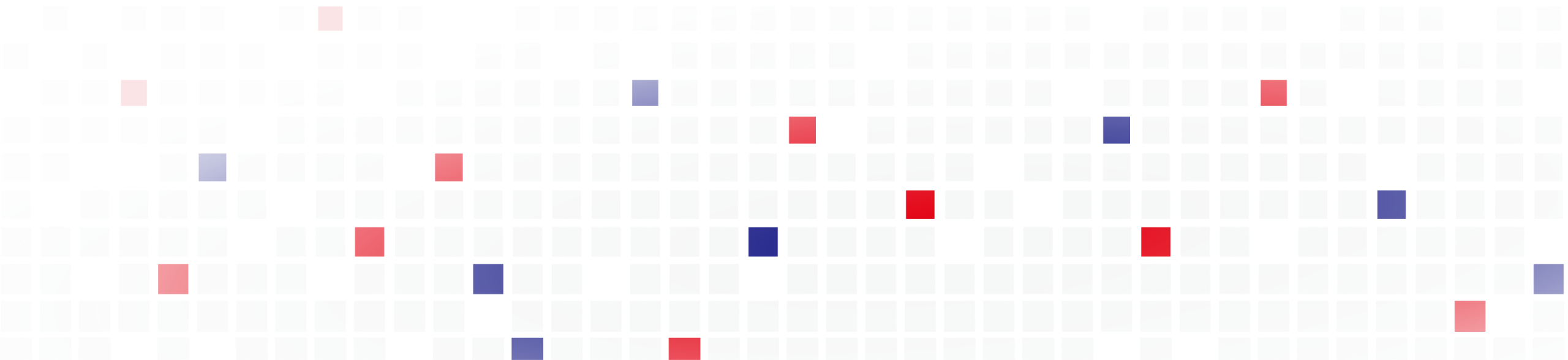
Software System Architecture

Application	 GNSS Location Tracking Multi-constellation positioning Real-time coordinate reporting	 LTE Cellular Connectivity LTE cloud connectivity HTTPS / MQTT data transport Offline data queue and retry	 BLE & RFID Local Configuration Local parameter configuration Short-range communication control	 CAN / Sensor data Acquisition CAN bus data acquisition G-sensor motion monitoring Analog input signal monitoring	 Status & Alerting Real-time status reporting Threshold-based alert triggers Tamper and anomaly detection
	 RTOS Scheduler FreeRTOS / Zephyr kernel Multi-task priority scheduling Interrupt and event handling	 GNSS Positioning Service NMEA sentence parsing Multi-constellation fusion Fix quality & accuracy	 Network Protocol Stack TCP/IP · UDP transport MQTT publish / subscribe	 Power Management Power-saving handling mode Deep-sleep wake-up scheduling Battery state-of-charge tracking	 OTA Firmware Update Signed firmware verification Rollback protection and recovery
Framework	 Peripheral Bus Manager CAN / SPI / I ² C / UART arbitration Event-driven message routing Bus conflict and timeout handling	 CAN Service CAN frame dispatch and filtering Application signal routing			
Driver	 LTE Modem Driver AT command processing Network registration & PDP RSSI / RSRP monitoring	 GNSS Receiver Driver NMEA stream parsing Cold / warm / hot start management Constellation selection	 BLE & RFID Radio Driver Connection management Device parameter configuration	 G-Sensor Driver 6-axis motion data acquisition Shock and impact event detection Orientation and tilt calculation	 Analog Input Driver ADC multi-channel sampling Signal conditioning and filtering Threshold comparator triggering
	 CAN Controller Driver CAN controller initialization Transmit and receive queue handling Frame error detection and recovery	 Flash Storage Driver SPI NOR flash read / write Wear leveling for log sectors Configuration partition management	 Power Management Driver Battery voltage & current monitoring Charging state machine control Low-battery shutdown protection		
HAL	 UART Interface LTE modem AT command path GNSS NMEA data stream Debug console and logging output	 I²C Interface Accelerometer register access RFID register access	 SPI Interface NOR flash storage interface High-speed peripheral access	 GPIO Interface Module enable and reset lines Interrupt and wake-up signals Status LED and indicator output	 ADC Interface Multi-channel analog input Voltage reference and scaling

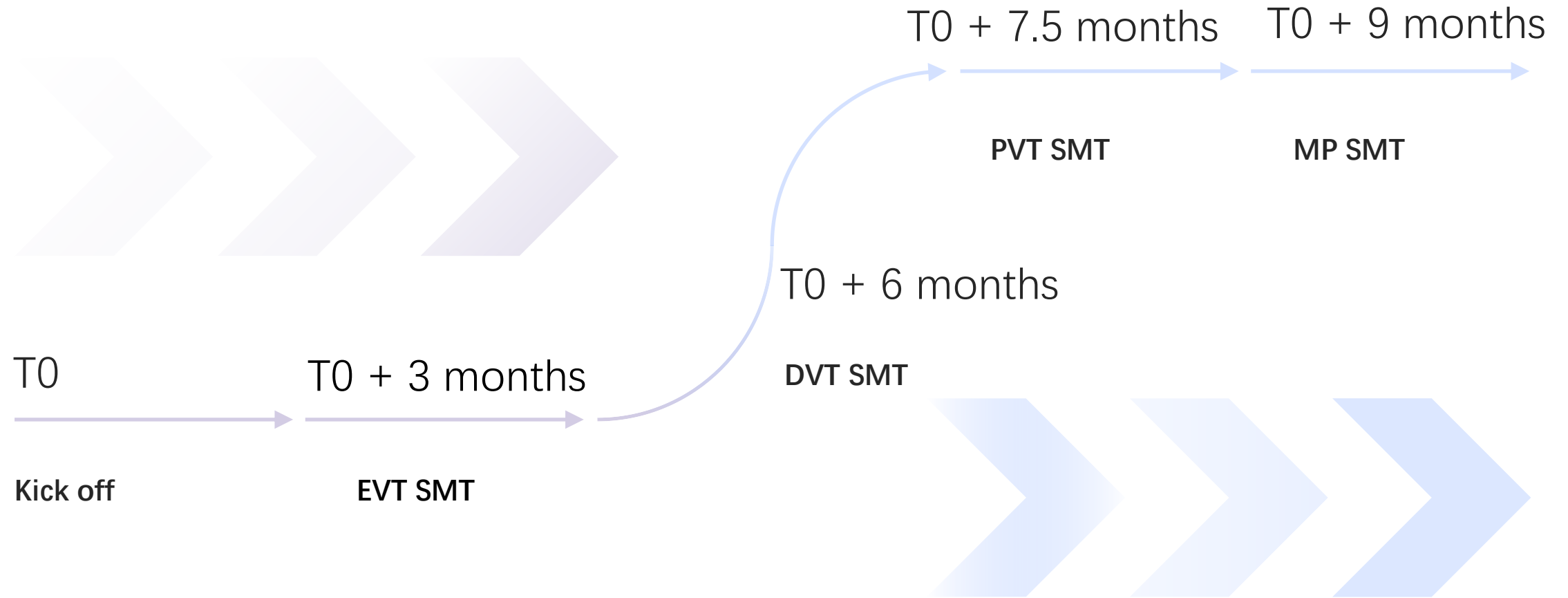
Software Feature list– Update

Device Software Feature list	Criteria
MCU related:	RTOS main application program design based on “EUP終端設備通訊協定”
	Bluetooth BLE/RS232 for device profile configuration
	Offline Resilience handling when LTE signal unavailable.
	Regional clock/time sync with cloud server
	RTC local timer support
	CAN J1939 data parsing program
	Customer cloud server communication management
	ADC and GPIO interrupt input detection program
Functional Components:	
G-Sensor	Motion, tilt, and vibration events monitoring through I2c control.
RFID	Identity, access tagging through I2c control.
Buzzer	Buzzer sound trigger through Triode with GPIO control.
Flash	Local data storage through SPI control
GNSS	GNSS location tracking through UART control
EG800Q-EU	Cellular data networking through UART control
External connectors control:	
* 1-Wire	Vehicle related sensor/actuator purpose
* 2*DIN(ACC/ALT0), 1*DIN(ALT1),1*DIN	Digital inputs for door sensor, ACC/ignition detect, smart box or alternate function pins
* DOUT1	Digital output for example relay, LED, buzzer driver
* ADC	ADC analogue signal for example fuel level, voltage sense
3*RS232	External CAN controller communication through UART
Power Management:	
System power management control	System sleep and run time power management handling
OTA & Updates	Firmware updates over-the-air via USIM

03 Project plan



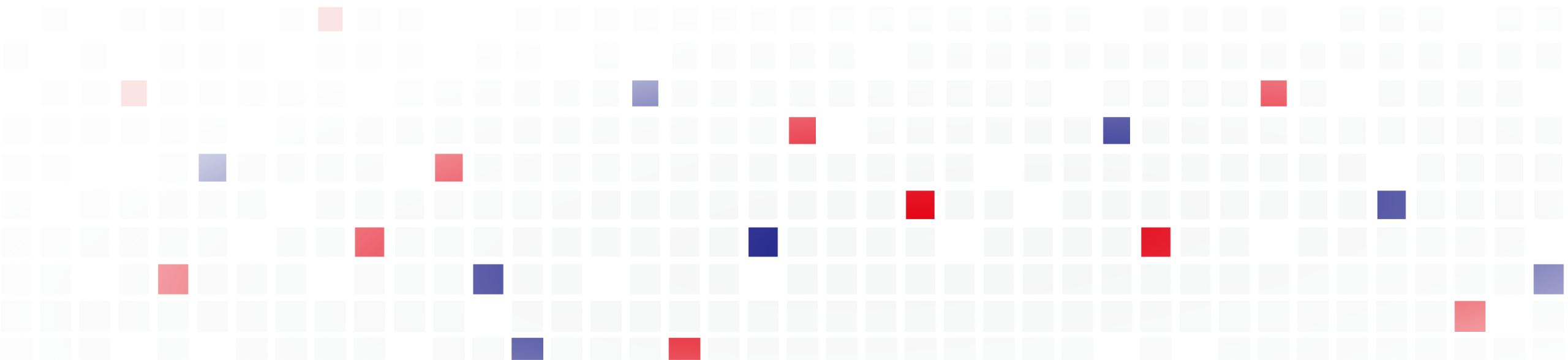
Development plan – Project schedule



Note:

- 1.The above is an estimated development plan. The specific timeframes will be determined after KO.
- 2.The target MP time is 2027.3.31. The specific adjustment will depend on the progress of the product certification and whether customer have any requirement change.

04 Quotation budgeting



Certification Details

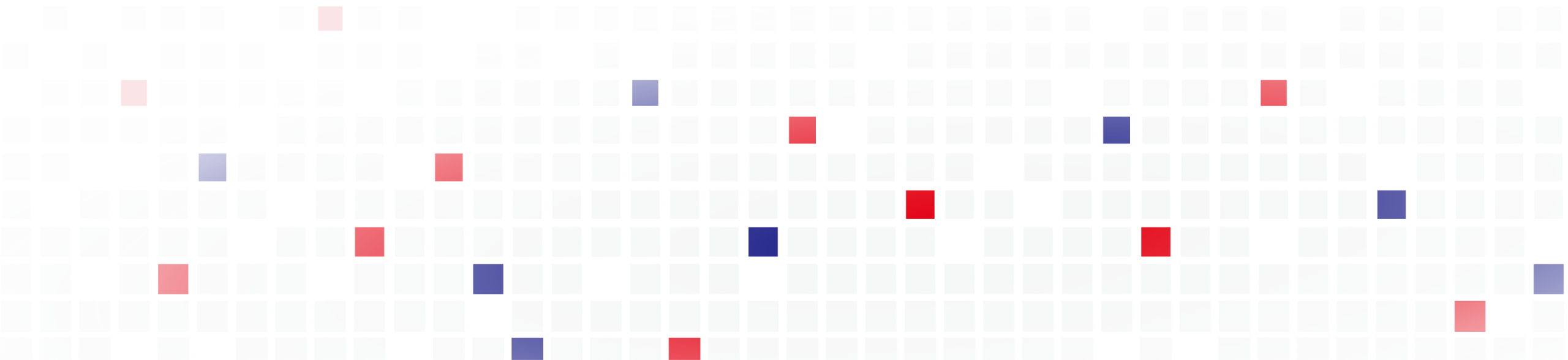
认证型号 Cert Model	产品类型 Product Type	产品规格 Product Spec	认证名称 Cert Name	认证范围 Cert Scope	认证费用 Cert Cost (USD)	认证周期 Cert Lead-time (Weeks)	认证人力 Esti. Cert Manpower	认证样品数 Cert Samples	备注 Note
TBD	Vehicle Tracker	include Quectel LTE EG800Q-EU (CE certified). include BT chipset, FR306x or nRF54LM20A.	MIC (QCVN standard)	1.QCVN 117:2023/BTTTT(RF) 2. QCVN 86:2025/BKHCN(RF EMC)	\$7,065	6-8 wks	0.4	4	1. It needs the actual importer hold the certificate. 2. Additional test cost would be required in case there is new test items or repeat test due to device design change or certification test failure. 3. According to QCVN31(which is a standard for power batteries), this is not necessary to do for tracker.
		LTE+2.4G BT+GNSS	CE(RED)	1.ETSI EN 300 328 2.ETSI EN 303 413 3.ETSI EN 301 908-1/13 4.EN IEC 62311 5.ETSI EN 301 489-1/ 17/19/52 6.EN62368	\$6,522	4-6 wks	0.3	6	1. The quotation excludes the outdoor safety test and SAR test. 2. Additional test cost would be required in case there is new test items or repeat test due to device design change or certification test failure.
			NBTC	/	\$3,261	6-8 wks	0.4	/	1. The quotation is based on CE(RED) certificate and report. 2. Thailand DLT: According to lab feedback, the central control box does not need to be implemented this certification; it is only required at the complete vehicle level.
			SIRIM	/	\$5,435	4-6 wks	0.4	1	1. It needs the actual importer hold the certificate. 2. The quotation is based on CE(RED) certificate and report.
			NCC	/	\$11,957	8 wks	0.4	8	1. It needs local test. 2. Additional test cost would be required in case there is new test items or repeat test due to device design change or certification test failure. 3. If the customer requests any hardware or mechanical changes after the design has been locked down, the customer shall bear any additional certification testing costs resulting from such changes.
			BSMI	/	\$3,261				
Total					\$37,501	Note: The total certified manpower investment is 1.9, which is belongs to the investment of the product line.			

Note:
1. 认证周期以及人力是参考预估， 实际最终是按照项目情况具体归集工时。
The certification lead-time and cert manpower are for reference estimates, and the actual final collection of working hours is based on the project situation.
2. 实验室费用/lab cost 报价1个月内有效， 最终以采购议价后的结果为准。
The cert lab cost is valid for 30 days, the final result will be based on the outcome of the procurement negotiation.
3. 请注意该报价不包含研发项目管理费用以及研发团队的人力投入成本。
Please note that this quotation excludes research and development (R&D) project management fees as well as R&D team labor costs. Kindly be advised.

Details of NRE for R&D & Production

Items				Unti Cost	Manpower	Total Cost
1	R&D	Hardware Development	Baseband Design (Schematic Design/PCB Layout/BOM)	\$4,700	3	\$117,500
2			RF Design (Impedance matching/RF tuning / Antenna Interface)			
3		EDA	Layout and related information package organization			
4		ATE	Factory Test Tool Development		1	
5		HW Testing	General HW\RF testing (EVT/DVT/PVT)		1.5	
			Reliability test			
6		Software Development	Device software functions development		9	
7		SW Testing	Device software full functions testing & stress testing		4	
8		Industrial Design	Industrial design of the device		1	
9		Mechanical Design	Mechanical design of the enclosure		2	
10	Production Management	Production	NPI&PC (Production Arrangement ,Material Allocation, Production Support)		1	
			PQE/SQE for Quality control		1	
11	PM	Project Management	Project Planning/Project Monitoring and Control /Risk Management		1.5	
12	Production	Pilot Run	3 times pilot-run, including: • production management/engineers • SMT stencil/jigs/etc. • Use of pilot-run SMT line / manual testing line	\$1,500	3	\$4,500
13		Pilot Run Equipment	• Testing fixtures • Accessories (cables, test SIMs, etc.) • Use of testing instruments	\$3,000	1	\$3,000
Total						\$125,000

05 Supplementary information



Milestones of deliveries

Milestone Deliverables		Quectel Responsibilities	Customer Responsibilities
T0	"T-Zero"	Prepare Project KO report & organize KO meeting with Customer	Sign the SOW Fulfil the Kick off NRE, attend the KO meeting and align technical details Provide mechanical constraints for the PCB Finalize and approve the Product PCB specifications
EVT	Engineering Samples	Delivery of Engineering Samples • Verification & QA test plan and test reports for EVT samples • SCH/PCB in PDF format for EVT build. • BOMs of main components and applicable datasheets • HW Test report for EVT build according to verification & QA test plan	Review the EVT Phase deliverables and do acceptance test and approve the deliverables
DVT	Design Validation Samples	DVT samples are functionally complete and should meet specifications	Review the DVT Phase deliverables and do acceptance test and approve the deliverables
PVT	Pilot Run	Delivery of PVT Test report & Samples • Verification & QA test plan and test reports for PVT samples • PCB in PDF format for PVT build • Bill of Materials and datasheets of main components (if there is update vs EVT) • HW Test report for PVT build according to verification & QA test plan	Review the PVT Phase deliverables and do acceptance test and acknowledge the deliverables
MP	Mass Production	Be prepared to accept Mass Production PO from customer	Send PO to Quectel according to the agreed pricing & MOQ & LT after certification done

Scope of Work Assignment

ODM Work	Quectel	Customer	Note
Industrial Design	Yes	/	• Need Customer review and confirmation
Mechanical Design	Yes	/	• Need Customer review and confirmation
Hardware Design & Manufacture	Yes	/	
Design Approval	/	Yes	• Need Customer review and confirmation
Device Software Development	Yes	/	• Standard CAN J1939 Quectel can develop
Hardware Test	Yes	Review	
Software Test	Yes	/	
EVT build	Yes	Review	• Customer shall approval samples within 30 days
DVT build	Yes	Review	• Customer shall approval samples within 30 days
PVT build	Yes	Review	• Customer shall approval samples within 30 days • PVT units are part of first MP order
NPI and ATE	Yes	/	
Quality Assurance	Yes	/	• Warranty/RMA follow up the master agreement
Certification	Yes	/	• Need Customer to confirm whether Quectel in charge of Certification
Software Tools	Yes	/	• Quectel can provide upgrade and debug tool
Documentation & Source Files	Yes	/	• Quectel can provide SCH(PDF version) and firmware

Note: Customer shall approve EVT and DVT samples within 30 days of delivery.

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HT Reliability Test Cases

Reliability Test Cases				
NO.	Classification	Test Items	Test Condition	Test Standard
1	Climate	低温运行	温度：-40℃ 时长：24H	IEC 60068-2-1 GB/T 2423.1
2		低温存储	温度：-40℃ 时长：48H	IEC 60068-2-1 GB/T 2423.1
3		高温存储	温度：85℃ 时长：48H	IEC 60068-2-2 GB/T 2423.2
4		高温运行	温度：85℃ 时长：24H	IEC 60068-2-2 GB/T 2423.2
5		高温运行寿命	依据产品寿命	
6		冷热冲击	温度：-40℃--+85℃ 24cycles	IEC60068-2-14 GB/T 2423.22
7		冷热冲击TTF	依据产品寿命	
8		温度循环	温度：低温-40℃；高温+85℃ 温度转转：5℃/min 24cycles	IEC60068-2-14 GB/T 2423.22
9		恒定湿热	85℃&85%RH 96H	IEC 60068-2-67 GB/T 2423.3
10		恒定湿热TTF	依据产品寿命	
11		交变湿热	1) 温度：25℃~+40℃ 2) 湿度：95%RH 3) 测试循环：2cycles亮度	IEC60068-2-30 GB/T 2423.4
12		盐雾	盐雾喷洒：中性盐雾浓度：5% 时间：24H；	IEC 60068-2-11 GB/T 2423.17
13		防尘	依据产品设计	IEC 60529 GB/T 4208
14		防水	依据产品设计	IEC 60529 GB/T 4208
15		氙气老化试验	1) 光源：0.55w/m2@340nm波段 2) 照射 20h, 停 照 4 h为一个循环，共4cycles	IEC60068-2-5 GBT 2423.24
16	Mechanical	随机振动	频率PSD范围 GRMS=27.1m/s2(=2.76g) 3方向*8H=24H	IEC 60068-2-64 GB/T 2423.12
17		机械冲击	波形：半正弦，加速度峰值：50G 持续时间：6ms 冲击方向：±X,±Y,±Z,18次	IEC 60068-2-27 ISO 16750-Part3
18		定向跌落	高度：1.0m 4角6面	IEC 60068-2-31 GB/T 2423.8
19		滚筒	高度：0.5m	IEC 60068-2-31 GB/T 2423.8
20	Structural	RCA耐摩擦试验	RCA纸带，500g	/
21		酒精耐磨试验	95%酒精，500g	/
22		铅笔硬度试验	HB,45°	/
23		挤压测试	100N	/
24		接口插拔试验	依据产品设计	/
25		按键寿命试验	依据产品设计	/
26	Battery	充放电寿命	300cycles（供应商）	/
27	Packaging	包装测试	ISTA 2A序列	/

Note: The above is Quectel test cases for device, just for reference, it can adjust according to customer requirement.

HT Reliability Test Cases

EMC Test Cases			
NO.	Test Items	Test Condition	Test Standard
1	ESD	根据产品定义或客户要求, 一般接触8KV, 空气12KV	IEC61000-4-2
2	Surge	根据产品定义或客户要求, 一般Vat±100V, Vbus±200V;	IEC61000-4-5
3	EFT	根据产品定义或客户要求(Only AC Power supply)	IEC 61000-4-4

Note: The above is Quectel test cases for device, just for reference, it can adjust according to customer requirement.

Contact information

Project Team Member List			
	Name	Position	Email
ODM Service	Leo Lin	Taiwan Local Product Manager	leopj.lin@pactw.com
	Kyson Zhang	Product Manager	kyson.zhang@quectel.com
	Rain Tian	Solution Manager	rain.tian@quectel.com
	Ethan Song	BB Engineer	Ethan.song@quectel.com
	Reuben.Bao	RF Engineer	reuben.bao@quectel.com
	Raoul Rao	Antenna Engineer	raoul.rao@quectel.com
	Shane Shang	ME Engineer	shane.shang@quectel.com
	William Wu	ID Engineer	william.wu@quectel.com
	Bill Yang	SW System Engineer	bill.yang@quectel.com
	Haned Kadrija	SW System Engineer	haned.kadrija@quectel.com

Thank you

For more information, please visit: [quectel.com](https://www.quectel.com), [LinkedIn](#), [Facebook](#) and [X](#).
Media contact: media@quectel.com

Sales support: sales@quectel.com
Technical support: support@quectel.com General: info@quectel.com

